

Research Article

Digital Twin-Enabled Delay Diagnosis Traceability and Propagation Process for Airport Flight Ground Service

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The emergence of digital twin technology offers a promising solution to address the limitations of traditional methods on early diagnosis and accurate propagation analysis of flight ground service delays. However, the application of digital twin technology in the civil aviation domain still stays at the lower maturity of the L2 level, which focuses on physical assets, operational data, and maintenance planning at airports, and failed to achieve the integration of flight ground operation mechanism and real-time data, making it difficult to realize timely delay diagnosis. The simulation model is also limited to the offline simulation technology, which cannot connect to real-time data for simulation from intermediate processes. In this work, we developed an advanced L3-level airport digital twin system for flight ground service processes delay diagnosis and propagation, which focused on real-time data-driven simulation models and machine learning applications to meet the timely and precision requirements. First, we used the Unity3D platform to construct static three-dimensional models of flight ground service objects on the airport cloud server. By parsing these behavioral state interfaces and mapping real-time dynamic data from the airport sensing and business systems, we achieved accurate visualization of the airport's dynamic operational processes. Then, a vehicle delay tree-based Bayesian diagnostic model was proposed in the digital twin system to analyze the relationships between multiple flights and service processes, which enables proactive diagnosis of the operation status and provides delay warning information. To improve the accuracy of propagation analysis, we proposed a “breakpoint” simulation method that enables real-time simulation starting from an intermediate moment, facilitating the inference of flight ground service delays since the early warning moment. In addition, two delay tracing and propagation algorithms were proposed to identify delays and investigate propagation paths. Leveraging real-time operational information, our approach provides valuable feedback for decision-making, empowering the airport manager to formulate precise optimization strategies. Experiments on real-world airport data have validated the effectiveness of our proposed method and provided practical recommendations for airport managers to reduce aircraft delays and improve airport operation efficiency.

Keywords: airport flight ground operation; Bayesian network; breakpoint simulation; diagnosis and propagation; digital twin; flight delay

1. Introduction

The airport's flight ground service process refers to a series of activities that are carried out between the arrival and departure of an aircraft, such as fueling, catering, and baggage

vehicle services [1]. Real-time flight ground service process delay diagnosis and propagation analysis have become increasingly crucial [2], as an abnormal or delayed service vehicle can not only cause delays for a single flight but may also propagate to other flights, resulting in delays in the

entire airport ground service network and reducing the efficiency of airport ground operations [3]. Today, the construction of virtual worlds through the dynamic information and models of systems evolving in real-time are known as the digital twin [4]. A distinguishing feature of the digital twins is the capability to be dynamically involved in real time and analyze and interpret intensive data flows from the physical world [5]. With the advent of sensor and IoT technology, real-time service time and location data of flight ground service processes can be obtained, and through the integration of corresponding business process models [6], the development of flight ground service digital twin systems has become possible. These digital twin systems connect the physical world operation with a digital representation, enabling real-time diagnosis of current flight ground service delays, as well as the prediction and control of future flight ground delay propagation. Therefore, the implementation of a digital twin system is imperative for achieving real-time flight ground service process diagnosis and propagation analysis, which is of great significance in improving airport operation efficiency.

Due to the significant development of data acquisition and automation technologies in the industrial manufacturing field, digital twin technology is popularly applied in equipment predictive maintenance [7], manufacturing process diagnosis and optimization [8], aerospace manufacturing, and health maintenance [9]. However, research on digital twin applications in airport operation areas started relatively late, with few progress made following the release of IATA's "VISION 2050" report in 2019, which highlighted the emergence of automated unmanned systems. The maturity of digital twin technology, as reviewed in the work in [10], encompasses four levels: L1 (predigital twin), L2 (digital twin), L3 (adaptive digital twin), and L4 (intelligent digital twin). According to this maturity level, research on digital twin technology in the airport field is at a relatively low L2 level, which is unable to meet the requirements for real-time data-driven implementation of flight ground service delay diagnosis and propagation analysis. Therefore, we need to further implement real-time updating of dynamic data mapping in the twin space and meet the L3 level of flight ground service digital twin system in real-time simulation models and machine learning applications. Specifically, we need to address the following three challenges:

1. Real-time data-driven diagnosis and warning of flight ground service process delays: How to effectively identify the interrelated complex network of multiple flight ground service processes in the entire airport and how to diagnose and provide early warning of flight delays caused by ground service process; these real-time warning moments serve as the basis for functional application in the twin system.
2. Real-time data and business mechanism integrated-based breakpoint simulation: How to integrate flight ground service business mechanism models with real-time data from the intermediate process of the delay warning moment, conduct real-time breakpoint simulation analysis in the digital twin virtual world, and obtain detailed data for the delays propagation analysis. The breakpoint refers to the simulation being based on the real-time warning moment, instead of traditional offline simulation from the initial moment, which may lead to significant deviations from actual operations.
3. Flight delay tracing and propagation analysis based on the twin data: How to identify the causes and propagation paths of flight delays from the twin data obtained by real-time breakpoint simulation and construct corresponding machine learning application models in the digital twin system, providing real-time feedback of information to the physical airport, thus providing effective assistance for decision-making in real-time airport operation management.

In order to solve the above challenges, we construct an L3-level digital twin system framework and apply it to the airport operation field for the first time. First, we use the Unity3D platform to build 3D models of flight ground service objects on the cloud server in the airport, parse the behavioral state interfaces of these objects, and synchronize the dynamic data collected by the airport sensing and business operation system. Then, we propose a dynamic Bayesian diagnosis model based on a vehicle delay tree and combine it with the real-time data in the digital twin system for early warning of delay diagnosis. In addition, to achieve an accurate simulation between the digital twin world and the physical operation process driven by real-time data, we propose a breakpoint simulation method that supports real-time simulation at intermediate time points, facilitating real-time decision-making. Finally, by leveraging real-time diagnosis and simulation data, we develop a flight ground service delay propagation analysis model in the digital twin system, providing real-time operational information for decision-making in real-time airport operation management.

Our main contributions are as follows:

- We propose a vehicle delay tree-based Bayesian network (VDT-BN) delay diagnosis model to address the issue of inadequate early diagnosis capability in existing methods. This method uses a Bayesian network (BN) to depict the fixed relationships within the flight ground service process and combines it with a vehicle association network to diagnose the external association relationship under the superposition of multiple flights, which greatly improves the accuracy of delay diagnosis.
- To address the inaccurate delay propagation analysis caused by the existing offline simulation method's inability to extrapolate from the intermediate moments based on real-time data, we propose a "breakpoint" real-time simulation method by modeling the physical process in a grid format parameter slicing, which enables the continuation of the breakpoint state at any time and improves the accuracy of real-time simulation. By performing real-time breakpoint

simulation at the early warning moment detected by the VDT-BN diagnostic model, the accuracy of flight delay propagation analysis is greatly improved.

- We developed two machine learning application modules that perform real-time traceability of delay causes and delay propagation analysis in the digital twin system. Experiments on real-world airport data demonstrate that the generated delay tracing and propagation path information significantly assists airport managers in formulating future operational plans and improving the efficiency of flight ground service operations.

2. Related Work

2.1. Flight Ground Service Delay Diagnosis and Propagation.

Over the past few decades, a variety of methods have been proposed to diagnose and optimize the operation of flight ground service processes. The aim of delay diagnosis and propagation control is to predict and analyze the operational conditions of departing flights during the ground service process, enabling proactive measures. Historical flight plans and resource data from service vehicles are utilized for forecasting and analyzing the operational process and providing timely delay alerts. The primary approaches employed include simulation model-driven methodologies and data-driven techniques.

Simulation methods utilize the operational parameters and service distribution patterns of aircraft, service vehicles, and ground operation rules to model physical operational processes. For example, multiagent simulation [11], discrete event simulation [12], Petri networks [13], and critical path analysis [14] are employed in the simulation modeling of ground service processes for flight operations. However, these methods rely on traditional offline simulations, starting from predefined inputs, and lack the capability to integrate real-time data from intermediate processes, leading to significant deviations between simulation results and real operations. Consequently, these methods are primarily used for macrolevel airport operation planning and resource control optimization, while facing challenges to meet the real-time and precision diagnosis requirements.

Data-driven methods utilize statistical analysis and causal mining of complex operational characteristics in flight ground service processes to diagnose or predict future delays. Real-time data analysis enables more timely monitoring and diagnosis of the actual airport operations. For example, delay propagation trees [15], BNs [16], spatial-temporal random forests [17], linear regression [18], and causal analysis [19] models are used for predicting and analyzing the propagation of flight delays. However, in real-time airport operations, multiple flights are simultaneously stacked together, forming a complex dynamic network relationship between service vehicles and aircraft. Data-driven methods face challenges in dynamically characterizing microlevel dependencies; therefore, existing research primarily focuses on macrolevel factors, such as flight delays caused by interairport route considerations or external factors such as air traffic control, weather, and passengers. At

the microlevel, only statistical analysis of the interaction patterns between a single airline and ground service processes has been achieved. As a result, current data-driven methods face challenges in accurately predicting and inferring the propagation of flight delays in airport ground operations.

In summary, the existing methods still encounter two significant gaps: insufficient capacity for early diagnosis of flight delays and imprecision in delay propagation analysis. Traditional simulation-based methods are constrained by an offline simulation approach, hindering the integration of real-time airport data and failing to meet the real-time and precision requirements for diagnosis. Data-driven approaches face challenges in capturing the intricate microlevel dynamic dependencies resulting from the overlapping of multiple flight ground service processes at airports, leading to difficulties in conducting accurate inference analysis for delay propagation. Therefore, it is crucial to seek new technical methods to solve the problems of the delays in diagnosing and propagation analysis of airport flight ground service during operation.

2.2. Digital Twin Technology and Application in Airport Operation.

In recent years, the advent of digital twin technology has provided a promising solution to address real-time delay diagnosis and propagation analysis problems in flight ground service processes. Initially conceptualized by Grieves [20], a digital twin is defined as a replicated representation of a physical object that integrates multiphysics, multiscale, and simulation-based monitoring and management throughout its life-cycle. With the advancement of Internet of Things technology, real-time sensor data from the physical environment are seamlessly mapped into the digital twin system [21]. This integration enables digital twins to accurately reflect the real-time operational status of the physical world and facilitate diagnostic capabilities. Furthermore, leveraging these data-driven insights, digital twins allow for simulation, prediction, and analysis of future scenarios within the physical domain [22].

Due to the advancements in data collection techniques and increasing automation trends in the industrial manufacturing domain, the digital twin techniques have witnessed rapid development [23, 24], such as predictive maintenance of industrial equipment [7], optimization of manufacturing process diagnostics [8], and aircraft manufacturing and health maintenance [9]. By leveraging the simulation capabilities of digital twins, proactive monitoring and identification of potential failures can be achieved before they occur [25]. With the development of airport information technology, the civil aviation field has also started the application of digital twin technology. However, early research efforts primarily focused on aircraft design and manufacturing [26], the research in civil airport operations beginning primarily in 2020. For instance, the International Air Transport Association (IATA) has outlined its “Vision 2050” document, signaling the gradual integration of automated unmanned systems for ground control activities in the foreseeable future [27]. Besides, the SITA Lab has proposed the development of a comprehensive

digital twin system tailored for utilization at a United States airport within the next 5–10 years [4].

The comprehensive review of digital twin techniques in this work [10, 28] presents a maturity classification framework consisting of four levels: L1 (predigital twin), L2 (digital twin), L3 (adaptive digital twin), and L4 (intelligent digital twin). Table 1 presents the extension of this maturity classification in the airport operation domain.

According to the above digital twin maturity levels, the current development in the airport operation domain is summarized in Table 2, and existing research still stays in the lower maturity levels of L1 and L2. For instance, in [31], digital twin technology is employed to characterize the physical operations of airports and integrate content management applications across various scenarios. In the context of airport infrastructure and 3D visualization modeling, the work in [32] presents a BIM-based airport asset model management platform, establishing a digital twin system that facilitates comprehensive life-cycle management of the airport's infrastructure and enables 3D visualization modeling. The work in [30] introduces an application scenario and conceptual framework for digital twins in airport ground traffic. Furthermore, the work in [29, 33] investigates the synchronization of air traffic data within a digital twin platform and explores secure data sharing. The authors of the work in [34] conduct an analysis of airport power network data and develop a corresponding digital twin platform for management purposes. However, these studies primarily concentrate on the conceptual aspects of digital twins at the L1 level, exploring real-time data and 3D models in airport operations, which lack practical application functionalities based on simulation models. Some researchers have made progress by incorporating simulation models or real-time data to construct digital twin applications at the L2 level. For instance, the authors of the work in [25, 27] propose a digital twin framework for airport ground traffic operations, enabling real-time monitoring and warning analysis of aircraft ground conflicts. Discrete modeling and mapping of aircraft ground operations data are performed. The work in [36] discusses the role of digital twins in the flight ground handling process, emphasizing the collaborative digitization of diverse data and object information and highlighting its impact on flight delay control from a collaborative perspective. Regarding baggage handling operations, the authors in [37] develop a three-dimensional visualization system based on process simulation models to visualize and validate the baggage handling process involving robotic handling systems and human operators. However, this system is primarily applied during the product design and development stages and lacks integration with real-time airport operation data. To address the real-time data integration problem, the authors of the work in [35] establish a digital twin system for aircraft cargo loading operations, enabling real-time visualization and optimization planning of loading equipment. However, as the cargo loading process is just one aspect of the flight ground service operation, achieving diagnostic and control capabilities for the entire flight ground service operation in the future will require the integration of additional business processes and real-time data.

Therefore, in the civil aviation airport domain, current digital twin applications are still concentrated at the lower maturity levels of L1 (asset model management and real-time data synchronization) and L2 (aircraft conflict detection and cargo handling control analysis). Consequently, effectively capturing the intricate business mechanisms underlying flight ground service operations remains a significant challenge. To enable real-time diagnostics and analyses of flight ground service processes, further implementation of L3-level digital twin applications is crucial. This involves integrating real-time data with simulation models, utilizing their descriptive capabilities to unravel the complexities underlying these processes. In addition, incorporating machine learning models enables real-time process diagnosis, planning, and control, providing valuable feedback to optimize physical airport operations.

3. Flight Ground Service Digital Twin System Framework

3.1. Flight Ground Service Scenarios and Problem Descriptions. The airport flight ground service process is completed by aircraft, various special vehicles, and operation managers. These vehicle resources, such as ferries, refueling, luggage, and food trucks are carried out in precise parallel or serial order. Besides, the aircraft also should accomplish several functions, such as loading the wheel gear, leaning on the ladder bridge, and withdrawing the wheel gear, within the time constraints. According to the standard flight ground service process requirement, the sequence of these service vehicles is shown in Figure 1(a), which is usually performed between the arrival and departure of the aircraft at the airport. In addition to the flight delays caused by weather-related issues or previous airports, the major factor at local airports is the inefficient service of special vehicle resources [38]. Therefore, what airports can control is the buffer time between flight arrival and departure, that is, to improve the accuracy and service efficiency of resource scheduling of various vehicles, absorb the delays caused before flight arrival, and use the buffer time to eliminate flight delays.

To depict the association relationship between multiple flights, we divide all flights into various intervals based on time interval T and then connect each flight according to the external service vehicles. As a result, these flights show a complex internet relationship with one another since the ground service process between multiple flights is affected not only by various vehicles within the aircraft but also by the other flights. As shown in Figure 1(b), the vehicle service resources of flight $FA_{W_1}^2$ are connected to flights $FA_{W_2}^1$ and $FA_{W_2}^2$. The fuel and baggage vehicles will serve flight $FA_{W_2}^1$ and $FA_{W_2}^2$ after servicing flight $FA_{W_1}^2$, and then they will continue to serve other aircraft in the $W_2, W_3, \dots$, and W_t intervals. Therefore, if the vehicle service on flight $FA_{W_1}^2$ is delayed, it will have an impact on a number of subsequent flights. In addition, new vehicle delays could occur during the propagation process, which would have the effect of superimposing one on top of the other, making this network process more complicated.

TABLE 1: Maturity level for airport digital twins extended based on [10].

Level	L1 predigital twin	L2 digital twin	L3 adaptive digital twin	L4 intelligent digital twin
Model sophistication	3D asset visualization technology	Virtual characterization of physical airports	Virtual characterization of physical airports with adaptive interfaces	Virtual characterization of physical airports with intelligent control feedback on physical airports
Physical twin	Does not exist	Exists	Exists	Exists
Real-time data mapping	Not applicable	Offline batch update	Real-time updates	Real-time updates
Real-time simulation model	Does not exist	No, offline simulation	Yes, real-time breakpoint simulation	Yes, real-time breakpoint simulation with intelligent interactive feedback
Machine learning applications	No	No	Yes, diagnosis, optimization, and feedback in twin worlds	Yes, diagnosis, optimization, feedback in the twin world and intelligent control of the physical world

TABLE 2: Digital twin applications in the aviation airport and their maturity levels.

References	Year	Research object	Purpose	Level
[29]	2020	Airport critical asset data	Represent the state and processes of the physical airport	L1
[30]	2020	Airport ground transport operation	Airport surface transportation digital twin framework for navigating operational challenges and planning for the future	L1
[31]	2021	Air traffic data	Air traffic safety controlling based on digital twin	L1
[32]	2022	Airport critical asset data	BIM-based modularized digital twin platform for optimal collection, integration, management, and utilization of airport critical asset data	L1
[33]	2022	Air traffic data	Serve as a resilient data platform, incorporating multiple data sources and enabling the interaction between an umbrella of twins	L1
[34]	2023	Airport electric data	A digital twin for the conversion of actual airports into the electric aviation era.	L1
[25]	2021	Airport surface traffic operation	A testbed for solutions in the field of centralized traffic control at airports	L2
[27]	2021	Airport surface traffic operation	Using a special simulation model to generate data streams of the traffic participants' location in the airport's transport network	L2
[35]	2021	Airport cargo operation	Real-time visualization and loading optimization operations of cargo assembly and security screening with digital twin	L2
[36]	2022	Airport ground operation	Shows the real state of the airport ground operation, combining virtual reality representations of the stands	L2
[37]	2022	Airport baggage handling process	Using process simulation platforms to visualize and validate baggage handling processes	L2

With the development of sensor technology, airports may be able to record the time and location information of various vehicles and aircraft by using video analysis and GPS, which provides effective data support for the operational diagnosis of flight ground service operations. However, due to the intricate linkage of the flight ground service network, knowing how to diagnose potential delay conditions in advance and determine the impact on future flights is the fundamental foundation for enhancing the efficiency of airport ground operating resources.

Therefore, the main scenario investigated in this work is that once a certain flight's service vehicle delivers a delay, how does it affect the subsequent flight service process and to what extent. Specifically, it includes (1) real-time delay diagnosis: modeling the correlation of multiple flight ground service processes at airports, diagnosing the real-time operation level, and evaluating whether subsequent flights are affected. (2) Delay propagation analysis: after identifying the delay on succeeding flights' ground service processes, it includes precisely depicting the propagation process and assessing the propagation pathways and networks.

3.2. Framework Structure of the Digital Twin Application System. In this work, we developed a L3-level digital twin framework in the airport flight ground service process to diagnose flight ground service delays and their propagation process. We use the Unity3D platform to model the physical airport in 3D visualization, parse the behavioral parameters of various physical objects, and use the airport's cloud server to store these twin models and present them visually. Figure 2 shows the framework's three main layers: data mapper layer, function execution layer, and application service layer. First, the data mapper layer gathers information about airport flight ground service vehicles, aircraft, and scheduling plans and performs real-time mapping of these physical airport operating data in the digital twin space. Then, we developed a VDT-BN delay diagnosis model in the function execution layer to depict the multiflight ground service network and diagnose the real-time operation status. Furthermore, we construct a real-time-based breakpoint simulation model, which connects the real-time data to execute an extrapolation simulation of the future operating state. Finally, we proposed a flight delay propagation analysis

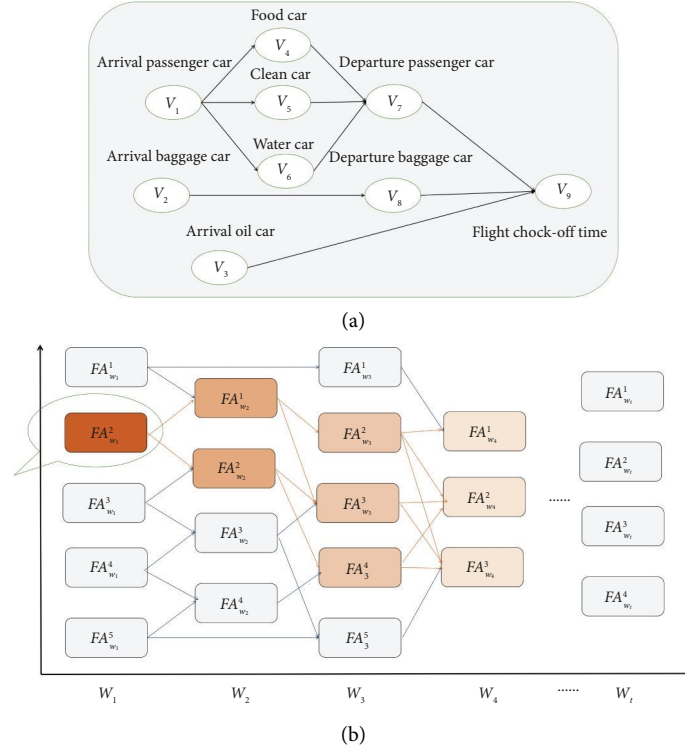


FIGURE 1: Multiflight ground service network process in the airport. (a) Single flight ground service process. (b) Multiflight ground service process.

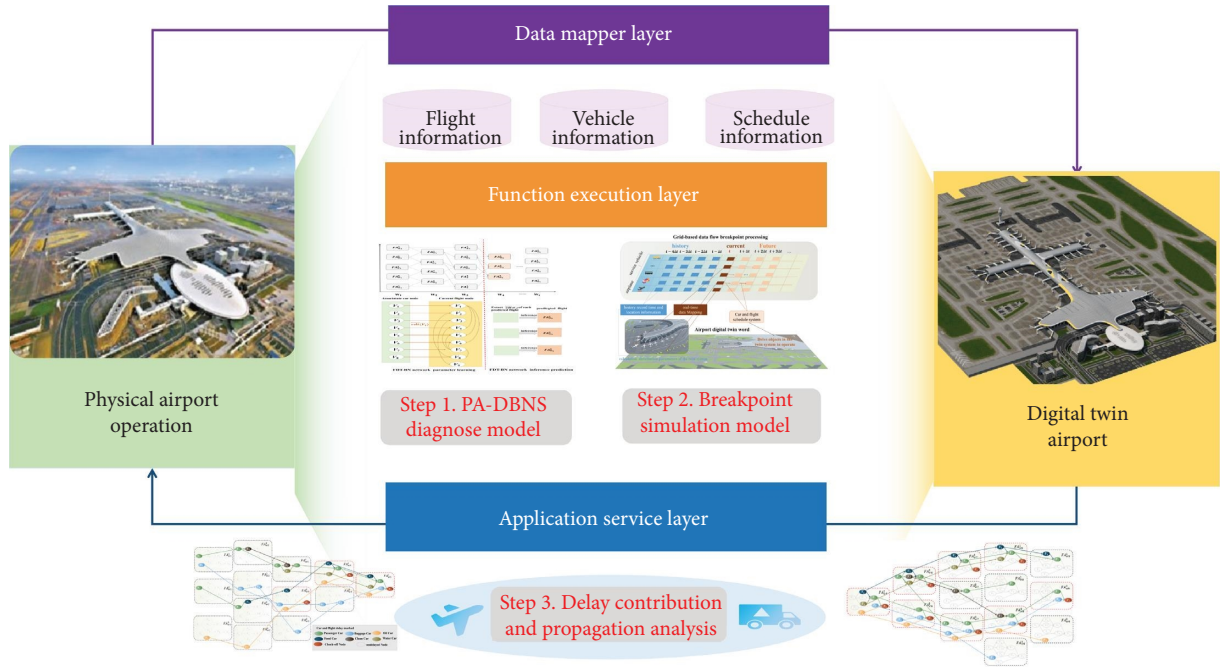


FIGURE 2: Digital twin system structure.

model in the application service layer that combines the diagnosis results with the breakpoint simulation model.

3.2.1. Data Mapper Layer. The physical process of airport flight ground service includes three types of objects: the vehicles used for services, the managers in charge of such vehicles, and

the aircraft themselves. As a result, we map the movement coordinates, time information, and future plan information of these objects in the data mapper layer. The physical space dataset that should be collected is summarized as follows:

$$D_{ps} = \{MI, TI, PI\}, \quad (1)$$

where MI, TI, PI stand for the movement, time, and future schedule plan information of the flight service vehicles. Since MI includes the position information $M_f = \{\text{loc}, \text{spd}\}$ and velocity information $M_C = \{\text{loc}, \text{spd}\}$ of aircraft and vehicles acquired from real-time sensors, $\text{loc} = \{\text{loc}_x, \text{loc}_y, \text{loc}_z\}$ represents mapped position information in the virtual space and spd represents velocity information used to drive aircraft and vehicles to synchronization move in the virtual world. As a result, MI can be defined as

$$\text{MI} = \{M_f, M_C\} \quad (2)$$

As shown in Figure 1, the flight ground service process includes passenger service, cleaning, and refueling node. We use the symbol V_i to represent the current vehicle node, and its values are dynamically updated according to real-time data transmission. t_i stands for the actual completion time of the relevant node, and if the operation is still running, t_i is denoted by *null*. Thus, TI can be defined as

$$\text{TI} = \{V_i, t_i\}. \quad (3)$$

The abbreviation PI stands for planned information and refers to details on the vehicles assigned to flights that have not yet taken place. According to Dispatch plan I, we use FA^j to represent a planned flight and $RA_j = \{RA_j^i\}$ to represent the various vehicles that are serving flight FA^j . Consequently, PI is defined as

$$\text{PI} = \{FA^j, RA_j\}. \quad (4)$$

3.2.2. Functional Execution Layer. The functional execution layer included the VDT-BN delay diagnosis model and the breakpoint simulation model, which were the foundation of the flight ground service digital twin framework. On the basis of historical data, VDT-BN first depicts the dynamic flight ground service network process, and real-time data are then used to analyze and diagnose each subsequent step in the future. The breakpoint simulation model, in contrast to conventional mathematical and simulation methods, is a real-time simulation model that is driven by business processes. We model the flight service process mechanism in 3D visualization software and connect the current real-time state to calculate and project the future flight ground security process. Therefore, the key core lies in how to interface with the ongoing service process and drive the objects in the 3D scene to move in conjunction with the future plan. In Sections 4.1 and 4.2, we will describe this layer in detail.

3.2.3. Application Service Layer. In the application layer, the functional layer's algorithms will be software-encapsulated and combined with the breakpoint simulation model to present the propagation process of future flight ground service delays and provide feedback to the physical operation airports to help managers make real-time decisions. To accomplish the analysis of the current delays causes and predict potential delays, we construct two reverse delay path traceability and a forward delay propagation algorithm. This

allows us to obtain a specific network path that will serve as the basis for the actual operation optimization. We will describe the details in Section 4.3.

3.2.4. The System Workflow. As shown in Figure 2, after defining the structure of the digital twin system, these three layers form an organic whole with a certain degree of versatility. The data mapping layer can incorporate additional data in the future, such as equipment information data and air traffic control information. The function execution layer can integrate other machine learning algorithms, for example, aircraft ground taxiing procedures, passenger check-in procedures, and baggage handling procedures. Finally, the application service layer calls real-time data and function execution and designs new application services, such as baggage and cargo fault detection, aircraft ground conflict detection, and air traffic control flow release. The entire system is driven by the application service layer, which selects the corresponding data and function execution, making the entire digital twin platform versatile. Different airports may have differences in the data mapping dictionary but we only need to parse their data, thereby ensuring the robustness of the entire platform. Different application services require designing the corresponding core algorithms at the function execution layer for functionality adaptation. Through the real-time data processing capability and the machine learning algorithm integration capability of the digital twin platform, the operational process of the physical airport can be effectively reflected in the virtual platform. This is the core architecture of the entire L3-level digital twin platform.

4. Research Methods and Models

4.1. VDT-BN Model for Delay Diagnosis and Warning

4.1.1. VDT-BN Construction

4.1.1.1. Definition of the BN Model. BN is denoted by $G = \{V, E, P\}$, where $V = \{v_1, \dots, v_a\}$ is a set of network nodes with a total of a variables. $E = \{(v_1, v_2), \dots, (v_i, v_j)\}$ stands for the edge between these nodes. The first node in (v_i, v_j) is represented by the parent v_i and the child is v_j . P is the probability distribution density in BN, which includes the distribution of the root node and conditional probability tables of other nodes. CPT represents the conditional distribution of child nodes given a joint parent, which can be expressed as $P\{v_j | \pi(v_j)\}$, where $\pi(v_j)$ stands for the parent node of v_j . Based on these features, we can use the business process in Figure 1 to depict the single flight ground service process, where each node represents a service vehicle and there are 9 nodes in total. However, the network consisting of multiple flight ground service processes was not a strict BN structure, as shown in Figure 3, and each flight was affected by the change of the inverted precedence node, and the impact of its change is closely related to the scheduling plan of various vehicles.

4.1.1.2. Definition of the VDT-BN Model. We developed a VDT-BN diagnostic model, which introduces vehicle delay propagation routes into the traditional BN structure. Each

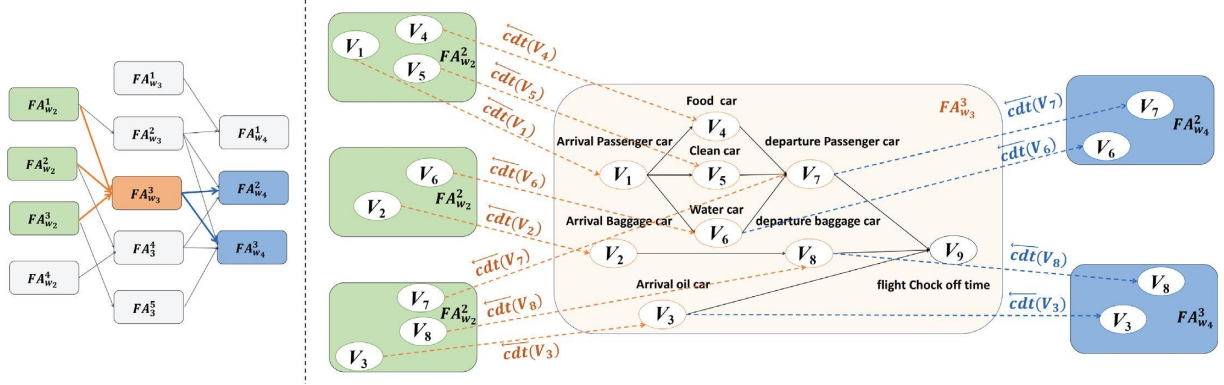


FIGURE 3: The relationship between multiple flights ground service process.

rectangle in Figure 3 represents a flight with a network of nine service nodes, and the flight's ground service time is impacted by the delays of six types of vehicles. $\overleftarrow{cdt}(v_i)$ denotes the vehicles associated with the antecedent sequence and $\overrightarrow{cdt}(v_i)$ denotes the vehicles subsequently associated with the v_i node. For instance, the prior sequence service nodes of flight $FA^3_{w_3}$ are distributed in three flights $FA^1_{w_2}$, $FA^2_{w_2}$, $FA^2_{w_2}$, and the nodes v_1 , v_4 , and v_5 of flight $FA^3_{w_3}$ are located in the prior sequence flight $FA^1_{w_2}$, while v_2 and v_6 are in $FA^2_{w_2}$. As a result, we incorporate BNs for defect and propagation diagnostics based on these prior association tree relationships.

4.1.1.3. Conditional Probabilities Definition. The conditional probabilities of each node were split into three scenarios: first node, middle node, and tail node. Equation (5) illustrates the conditional probability function for the first node, which is influenced by the associated vehicle precedence node $\overleftarrow{cdt}(v_1)$ in the preceding order. Equation (6) illustrates the probability function for the middle node, where the delay time for the chock-off node is influenced by the internal nodes, and equation (7) illustrates the conditional probability function for the tail node.

$$p(v_1 | \overleftarrow{cdt}(v_1)) = \frac{p(v_1, \overleftarrow{cdt}(v_1))}{p(v_1)}, \quad i = 1, 2, 3, \quad (5)$$

$$p(v_i, \overleftarrow{cdt}(v_i), \pi(v_i)) = p\left(\frac{\overleftarrow{cdt}(v_i), \pi(v_i)}{p(\pi(v_i))}\right), \quad i = 4, 5, 6, 7, 8, \quad (6)$$

$$p(v_9 | \pi(v_9)) = \frac{p(v_9, \pi(v_9))}{p(\pi(v_9))}. \quad (7)$$

4.1.2. Parametric Learning and Belief Propagation Inference. Since each flight is affected by different proceeding flight service nodes, we need to first rebuild the original data according to the vehicle delay tree to obtain the network structure of VDT-BN and then learn its parameters. Then, inferential forecasts for upcoming flight ground services are made using the learned parameters. Using the equal-width method, we discrete the data into 5 zones [0,5,10,15,30], which correspond to [1–5] delay levels, where Level 0 corresponds to no delay, Level 5 corresponds to

delays longer than 30 min, and Levels 2, 3, and 4 correspond to the remaining three interval contents, respectively.

4.1.2.1. Rebuilding the Original Data. Since the prior-order association of each flight is distributed across different flights, the network structure is fixed, as shown in Figure 4. The prior-order network connection points consonantly include V_1 to V_8 for each flight ground service network. We reconstruct the original data based on the invariant nature of this structure, extract the vehicle nodes of the prior-order

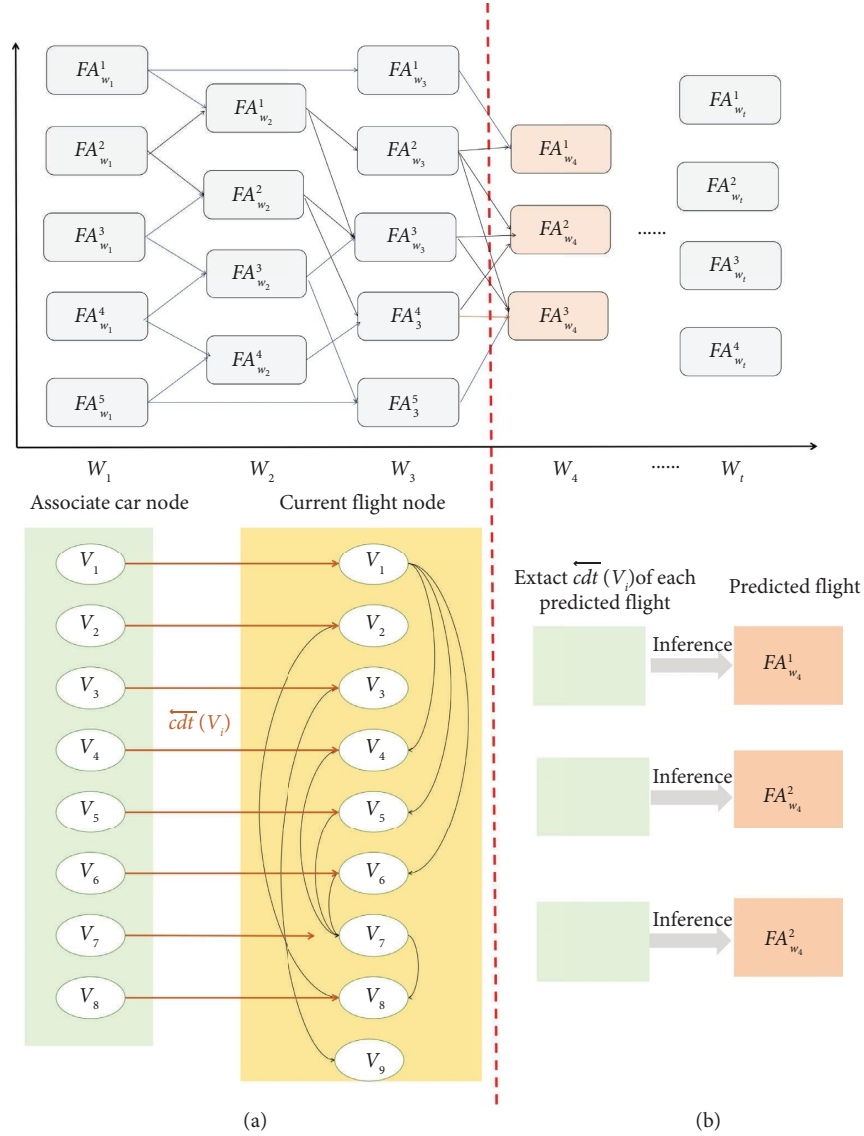


FIGURE 4: VDT-BN model parameter learning and diagnosis inference. (a) VDT-BN network parameter learning. (b) VDT-BN network inference prediction.

association of each flight at the current time, and then construct the VDT-BN data.

4.1.2.2. VDT-BN Network Parameter Learning. Assuming that the network hyperparameter of the datasets $D = \{V_1, V_2, \dots, V_a\}$ is Θ , we use θ_{ijk} to denote the j th combination of the parent node $\pi(v_i)$ and associated node $\overleftarrow{cdt}(v_i)$ of v_i , and the i th node takes the value of k parameters. i.e., $\theta_{ijk} = p(v_i = k | \overleftarrow{cdt}(v_i), \pi(v_i))$, where r^i denote the state number node v_i and q_i is the total number of combined states of $(\overleftarrow{cdt}(v_i), \pi(v_i))$. We introduce α_{ijk} as the Dirichlet distribution, which stands for the prior distribution of θ_{ijk} . The parameters' learning was transformed into the maximum posterior estimate (MAP) problem, and its objective function can be expressed as equation (8), where c

is a constant and N_{ijk} stands for the statistic at the i th combination of the parent nodes of v_i in dataset D . Equation (9) was used to get the MAP estimate after maximizing equation (8). As a result, the process for dataset reconstruction and parameter learning is shown in Algorithm 1.

$$\log P(\theta | D) = \sum_{i=1}^n \sum_{j=1}^{q_i} \sum_{k=1}^{r_i} (\alpha_{ijk} + N_{ijk}) \log \theta_{ijk} + c, \quad (8)$$

$$\tilde{\theta}_{ijk} = \frac{N_{ijk} + \alpha_{ijk}}{\sum_k (N_{ijk} + \alpha_{ijk})}, \quad (9)$$

4.1.2.3. Inference of the VDT-BN Model. We perform dynamic belief propagation based on the vehicles and flight schedule information. For the flights $\{FA^1_{w_t}, FA^2_{w_t}, \dots, FA^j_{w_t}\}$

Input: Flight ground service dataset D and scheduling plan for flights and service vehicles P

Output: VDT-BN model hyperparameter Θ and its conditional probability distribution CPT

1. According to the flight ground service process in Figure 1, parse the dataset D and construct the nodes and edges in the BN G
2. Based on the scheduling plan P and the associated flight nodes $\overleftarrow{cdt}(v_i)$ of all nodes v_i in network G , we construct a new VDT-BN network G' as shown in Figure 4 and reorganize the datasets
3. Combining the structural relationships in the new network G' , the conditional probability distribution function is constructed using equations (5)–(7)
4. Learning the distribution parameter Θ using equations (8) and (9)
5. The best hyperparameters of VDT-BN are obtained

ALGORITHM 1: VDT-BN model parameter learning algorithm.

in the interval w_t , the associate node $\overleftarrow{cdt}(v_i)$ of the v_i in each flight $FA_{w_t}^j$ were sequentially extracted and then permuted to create a new dataset. Following this, the flights in w_t are inferred using the sequential order. We use equations (5)–(7) to calculate the joint probability distribution from the first to the last and choose the distribution with the highest probability as the inference result. Taking flight $FA_{w_t}^1$ as an example, the inference result of its node v_1 is $v_1(FA_{w_t}^1) = \max(P_{FA_{w_t}^1}(v_1 | \overleftarrow{cdt}(v_1)))$, and the details of the process are shown in Algorithm 2.

4.1.3. Real-Time Diagnostic Warning and Propagation Analysis Metric. We built a set of diagnostic indexes for early warning analysis, so that effective delay propagation analysis can be performed in the digital twin system. There are 2 indicators used for real-time diagnostic, the first is the service vehicle real-time delays Dc_{w_t} in the current w_t interval, with $dc_{FA_{w_t}^j}^i$ standing for the i th vehicle delays time in flight $FA_{w_t}^j$. The Dc_{w_t} can be expressed by using the following equation:

$$Dc_{w_t} = \frac{1}{m} * \sum_{i=1}^m \sum_{j=1}^{q_i} \frac{1}{q_i} * dc_{FA_{w_t}^j}^i, \quad (10)$$

where q_i denotes the number of delays sent by the i th service vehicle in w_t , $\sum_{j=1}^{q_i} 1/q_i * dc_{FA_{w_t}^j}^i$ denotes the average delay time of that type of vehicle, and there are m types in total.

The second metric is the average delay time Df_{w_t} of the delayed flights in interval w_t , and $df_{FA_{w_t}^j}$ is used to denote the delay time of flight $FA_{w_t}^j$, and there are a total of r_i flights in w_t , so Df_{w_t} can be expressed as

$$Df_{w_t} = \frac{1}{r_i} * \sum_{j=1}^{r_i} df_{FA_{w_t}^j}. \quad (11)$$

Algorithm 3 depicts the steps in this diagnostic process. By repeatedly calculating the historical intervals, we get the associated diagnostic indicator thresholds. Then, utilize the VDT-BN model to predict future delays and contrast them with actual values to determine the appropriate minimum thresholds.

4.2. Grid Segment-Based Real-Time Breakpoint Simulation Model

4.2.1. Grid State Parameter Extraction. As shown in Figure 5, we create a 3D model based on airport CAD designs and ground service procedures. We build corresponding intelligent models by analyzing parameters from airport historical data, such as the service path of aircraft and vehicles, and the operating procedure between them.

4.2.1.1. Taxiing Path Analytical. $Froad_{FA_{w_t}^j}$ is a representation of the aircraft's taxiing path. According to the airport's operational procedures, each aircraft's taxiing path is determined by the runway number $L_{FA_{w_t}^j}$ and the aircraft stand $S_{FA_{w_t}^j}$, and equation (12) was used to analyze the path of the aircraft. Likewise, the service route of each service vehicle $Croad_{V_i}$ is determined by the vehicle's origin and destination, with $Cloc_i^s$ denoting the vehicle's starting point and $Cloc_i^e$ denoting the destination of the i th vehicle. We fix the start and end point, each parking space, and the fixed position of the vehicle at the airport's special location to make the computation simpler. This makes it possible to determine the overall limited service path, and the analytical result is stated by the following equation:

$$Froad_{FA_{w_t}^j} = f(L_{FA_{w_t}^j}^j, S_{FA_{w_t}^j}^j), \quad (12)$$

$$Croad_{V_i} = f(Cloc_i^s, Cloc_i^e). \quad (13)$$

4.2.1.2. Vehicle Service Time Analysis. After obtaining the service paths of various service vehicles, we also need to analyze the service time distribution of each vehicle for the subsequent simulation process. According to the existing work [14], we mainly choose five commonly used distribution functions: normal distribution, gamma distribution, beta distribution, Exponpow distribution, and Cauchy distribution for fitting, which are defined in the following equations:

Input: Historical dataset D , future flight and service vehicle plan P , and learned VDT-BN model parameters CPT and Θ

Output: The predicted dataset D' of flight ground service nodes in future intervals

1. According to the flight ground service process in Figure 1, parse the dataset D and construct the nodes and edges in the BN G
2. For each flight F in W , extract the association nodes $\overleftarrow{cdt}(v_i)$ according to the vehicle scheduling plan P and reconstruct the VDT-BN dataset
3. According to the VDT-BN network structure G' and model parameters CPT and Θ , the delay probability distribution of each node v_i in flight FA_{wt}^j is calculated sequentially using equations (5)–(7), and the largest probability is selected as the predicted result
4. Repeat Steps 2–3 until all flights in the interval are calculated and determine whether the V9 node(chock-off time) in all flights in the interval has no delays
5. If v9 node delayed in w_t , continue to calculate the flight in w_{t+1} , until all flights in w_{t+1} are not delayed

ALGORITHM 2: VDT-BN model belief propagation algorithm.

Input: Real-time diagnostic metric calculation DC_{w_t} and Df_{w_t}

Output: diagnostic metric thresholds

1. Initialize the maximum (most stringent) diagnostic threshold values based on the real-time DC_{w_t} and Df_{w_t} , where all conditions are initially diagnosed as delays
2. Determine if the current values exceed the threshold. If not, recalculate DC_{w_t} and Df_{w_t} , and return to Step 1. If they do exceed the threshold, proceed to Step 3
3. Apply the VDT-BN algorithm for prediction to obtain a predicted value
4. Compare the predicted value with the actual value to determine the diagnostic accuracy. If the accuracy meets the required standards, output the diagnostic threshold. If not, proceed to Step 5
5. Gradually reduce and adjust the diagnostic threshold to decrease sensitivity. Adjust the threshold based on the diagnostic accuracy until the required precision is achieved
6. Once the precision requirements are met, finalize the diagnostic threshold

ALGORITHM 3: Diagnosis index threshold analysis algorithm.

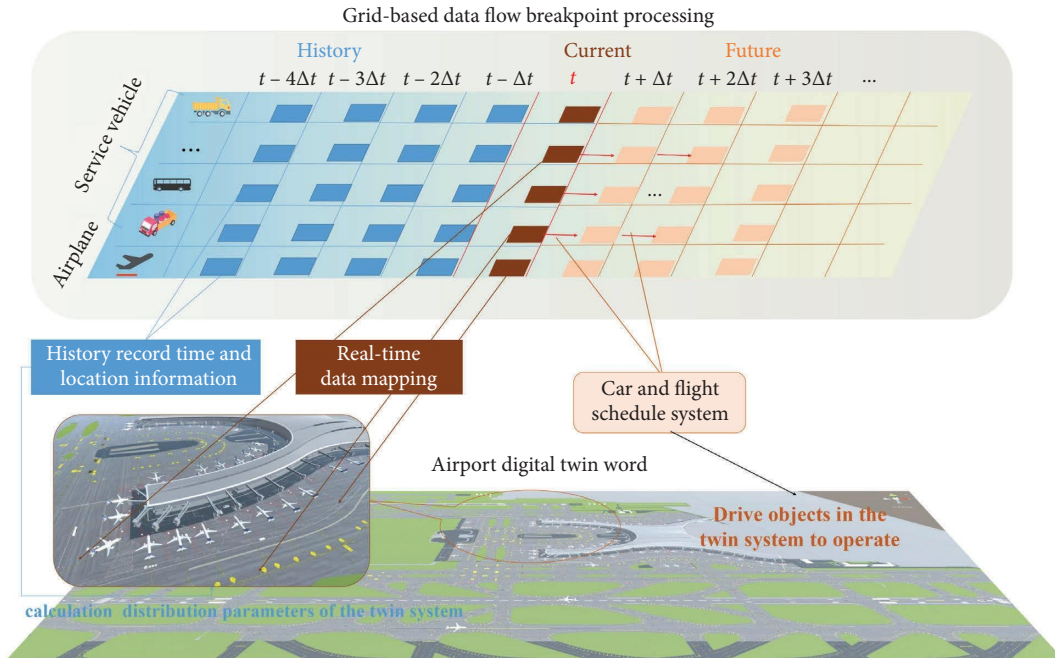


FIGURE 5: Grid-based breakpoint simulation flow.

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty, \quad (14)$$

$$f(x, \beta, \alpha) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0, \quad (15)$$

$$\Gamma(\alpha) = \int_0^{+\infty} e^{-x} x^{\alpha-1} dx (\alpha > 0),$$

$$f(x) = \left(e^{1-e^{\lambda x}} \right) e^{\lambda x^\kappa} \lambda \kappa x^{\kappa-1}, \quad 0 < x, \quad (16)$$

$$f(x) = \frac{1}{\pi} \frac{b}{b^2 + (x-a)^2}, \quad (17)$$

$$f(x; \alpha, \beta) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad (18)$$

$$B(\alpha, \beta) = \int_0^1 t^{\alpha-1} (1-t)^{\beta-1} dt,$$

where the normal distribution parameter in equation (14) is μ and the scale parameter is σ ; the shape parameter in gamma distribution of equation (15) is α , which allows the distribution to represent many more distribution shapes, and β is the scale parameter. The shape parameter in the Exponpow distribution of equation (16) is K . In the Cauchy distribution defined by equation (17), a is the location parameter and b is the scale parameter. Finally, the beta distribution in equation (18) is a continuous probability density distribution, where the shape parameter is determined by two parameters a and b . In addition, four additional distributions: the chi-square distribution, t-distribution, power-law distribution, and Laplacian distribution, were chosen for the comparative analysis of the validity of various node service time distributions.

4.2.1.3. Fitting Evaluation. We choose the Kullback–Leibler divergence to evaluate the fitting performance and quantify the difference between these distributions. Equation (19) can be used to express KL divergence. It assumes that the real flight ground service time probability distribution is $P(x)$ and the approximate probability distribution is $Q(x)$.

$$D_{KL}(P \| Q) = \mathbb{E}_{x \sim P} \left[\log \frac{P(x)}{Q(x)} \right] = \mathbb{E}_{x \sim P} [\log P(x) - \log Q(x)]. \quad (19)$$

4.2.2. Real-Time Data-Based Breakpoint Simulation. We assume that the seat of flight $FA_{w_i}^j$ is $S_{FA_{w_i}^j}$, the runway is $L_{FA_{w_i}^j}$, and $Cloc_i^t(FA_{w_i}^j)$ represents the i th vehicle's current position information, where $FA_{w_i}^j$ represents the vehicle's service flight and $Cloc_i^s(FA_{w_i}^j)$ represents the starting position of the vehicle. For vehicles and aircraft in service, the process of path continuation at the breakpoint time is as follows.

4.2.2.1. Path Information Analysis. By backtracking from the breakpoint moment, it is possible to determine the

vehicle's original departure position information. Next, equation (20) can be used to determine the vehicle's service path at the breakpoint moment. Similarly, according to the current position of the aircraft, its aircraft position can be obtained as well as the runway numbers $L_{FA_{w_i}^j}$, $S_{FA_{w_i}^j}$, and the taxi path information $Froad_{FA_{w_i}^j}$ can be obtained by using equation (12). Finally, the aircraft and vehicles are driven to complete the subsequent simulation based on the location data at the breakpoint instant. Besides, using the starting position data and equations (12) and (13), the path information for vehicles and airplanes that have not yet entered service or have finished service is directly calculated, and these items are then driven to move in the simulation platform.

$$croad_{V_i} = f\left(Cloc_i^s(FA_{w_i}^j), S_{FA_{w_i}^j}\right), \quad (20)$$

$$Cbeg_i(FA_{w_i}^j) = \text{Min}(Cend_{\pi_i}(FA_{w_i}^j)), \quad (21)$$

$$Cend_i(FA_{w_i}^j) = Cbeg_i(FA_{w_i}^j) + Cser_i. \quad (22)$$

4.2.2.2. Assessment of the Process and Time. According to the service process in Figure 1, each type of service vehicle has an arrival time, service start time, and end time, which are denoted by $Cavt_i(FA_{w_i}^j)$, $Cbeg_i(FA_{w_i}^j)$, and $Cend_i(FA_{w_i}^j)$, respectively. Besides, the operation of each vehicle needs to wait for the end of the preorder before it can be carried out. The judgment rule is as shown in equation (21), where $Cavt_{\pi_i}(FA_{w_i}^j)$ represents the completion time of the presequence service node. For example, node v_7 represents the passenger vehicle for the arrival flight, and the service can only start after the completion of the preorder V_4 , V_5 , and V_6 nodes. The end time $Cend_i(FA_{w_i}^j)$ of each service vehicle is shown in equation (22), which can be calculated by adding the service start time $Cbeg_i(FA_{w_i}^j)$ to the service duration $Cser_i$, where $Cser_i$ is estimated based on the parameter distribution in historical data.

4.2.2.3. Real-Time Parameter Analysis. According to the different situations of the above breakpoint process, the corresponding real-time parameters can be obtained according to the breakpoint simulation method, and then the simulation platform is used for prediction. The process is shown in Algorithm 4.

4.3. Delay Traceability and Propagation Analysis Module in the Digital Twin System

4.3.1. Delay Reverse Traceability Analysis. Reverse traceability is performed when a delay is sent at the chock-off node in the ground service process of flight $FA_{w_i}^j$, which includes node traceability within the flight and association traceability between the flights. While the latter depends on the vehicle delay tree $cdt(v_i)$, the former depends on the parent node $\pi(v_i)$.

Input: Current real-time data and information on future plans for various vehicles and aircraft $D_{ps} = \{MI, TI, PI\}$

Output: Complete the simulation of future plans for flights and output the node times for all service vehicles and aircraft

1. Mapping the location information MI of vehicles and aircraft based on real-time data and resolving the runway information $L_{FA_{wt}^j}$ and aircraft stand information $S_{FA_{wt}^j}$
2. The simulation objects are initialized, the associated virtual objects are formed based on the ground service process, and vehicles and airplanes with unstarted or completed services are selected
3. For the vehicles and aircraft in service, the future service paths $Croad_{v_i}$, $Froad_{FA_{wt}^j}$ at the breakpoint moment are resolved using equations (12) and (20), the breakpoint location information is mapped in the path, and the aircraft and vehicles are driven to the target point from the breakpoint moment
4. Equations (21) and (22) are used to determine the start of service time $Cbeg_i(FA_{wt}^j)$ of various service vehicles in turn, and the vehicle service time $Cser_i$ is calculated based on the status information of the vehicle and the business process-driven simulation model is applied
5. Repeat Steps 2–4 until all service processes for all flight service nodes are completed and record the completion time for each node $Cbeg_i(FA_{wt}^j)$

ALGORITHM 4: Grid-based break point simulation algorithm.

4.3.1.1. Delay Nodes Search. We find the delays of the service nodes within flight FA_{wt}^j , trace its associated nodes from backward to forward according to the flight ground service process, and identify the causes of delays. We start from the last node V_9 of flight FA_{wt}^j , trace backwards to find its parent node $\pi(v_i)$, judge whether it is delayed, and continue tracing back to its parent node if it is delayed until it stops when there is no delay. The set of all nodes identified is represented by VS_0 . As an illustration, $VS_0 = \{V_7, V_8, V_4, V_5\}$, the parent nodes of V_9 were V_3, V_7, V_8 ; we identify V_7, V_8 delay, then the parent node of V_7 is V_4, V_5, V_6 ; we identify the delay V_4, V_5 , and iterate until the first node V_7 , and stop if no delay occurs.

4.3.1.2. Propagation Path Analysis. We continue our search for associations between flights after obtaining the propagation path VS_0 within one flight and use the vehicle delay tree to judge the previous flight node $\overleftarrow{cdt}(v_i)$ of each point V_i in VS_0 . Then, we set $\overleftarrow{cdt}(v_i)$ as the root node to begin the search, the judgment within the flight to get the propagation point set V_0^i , and finally get all the points of the antecedent flight association set $VS_1 = \{VS_0^k | V_k \in VS_0\}$, where VS_0^k indicates that the node starts from VS_0 to search for its antecedent $\overleftarrow{cdt}(v_i)$ in the flight where the associated delay node is set, and after searching for all nodes in VS_0 , the first level of associated delay set VS_1 is obtained. The above process is repeated, starting from each node VS_0^k in VS_1 , and the antecedent flight is searched to get VS_2 , until the node of antecedent flight is not delayed, then backtracking is stopped, and finally the propagation search result of reverse sourcing $VS = \{VS_0, VS_1, VS_2, \dots, VS_p\}$ is obtained, where VS_p denotes the final set of source-cause nodes. Algorithm 5' process is shown as follows.

4.3.2. Delay Forward Propagation Analysis. Forward propagation, in contrast to delay reverse traceability analysis of the previous operation process, is the projection of the

future operation process through the breakpoint simulation model after the flight delay occurs. After obtaining the detailed dataset by the digital twin system, we can analyze the propagation impact of the currently delayed flight on the subsequent flights and obtain the detailed propagation paths of the flight ground service delays, which can effectively help managers make specific optimization strategies in real time.

First, we identify in flight's internal service nodes that contribute to the chock-off node v_9 delay. Similar to Step 1 in Algorithm 4, we calculate the propagation results within one flight, get the subsequent associated nodes of each node v_k based on the recognized vehicle association network, represented by $\overrightarrow{cdt}(v_k)$, and get the association set VS_0 . Then, we explore the association from $\overrightarrow{cdt}(v_k)$ and evaluate their delays up until the last node. Finally, similar to the traceability algorithm, we calculate the associated flights from VS_1 by iterating until none of the subsequent flight nodes are delayed. This gives us the propagation search result of reverse path $VS = \{VS_0, VS_1, VS_2, \dots, VS_p\}$, where VS_p denotes the set of nodes of flights whose delays are ultimately eliminated. Algorithm 6 illustrates these processes.

4.4. Research Data and Experimental Setup

4.4.1. Research Data. In this work, the datasets were collected from the Chongqing Jiangbei International Airport's T3A Terminal between July 1 and November 30, 2018. The flight times are from 9:00 a.m. to 23:00 p.m. There are an average of 186 days of flight ground service data, and a total of 6 types of service vehicles, including passenger, oil, clean, food, baggage, and water vehicles, are chosen for verification analysis. The original flight datasets sample is shown in Table 3, and the service time data of these vehicles can be directly obtained from the airport A-CDM system collected by GPS, video analysis, and other technologies. The A-CDM system also keeps track of the original service time for different vehicles, and based on this information, we can create the grid-base simulation parameters and real-time mapping interfaces for the digital twin system.

Input: Node service time of flight $FA_{w_i}^j$ whose chock-off node is delayed (i.e., v_9 is delayed)

Output: The set $VS = \{VS_0, VS_1, VS_2, \dots, VS_p\}$ of reverse delay traceability nodes starting from flight $FA_{w_i}^j$

1. Search inside the flight: starting with node v_9 of flight $FA_{w_i}^j$, identify its parent nodes v_3, v_7 and v_8 to filter out the delayed nodes, and keep doing so until the search reaches the root nodes v_1 , and obtain the internal node association set VS_0
2. Search between flights: seeking the node $\overleftarrow{cdt}(v_i)$ in the postorder-related flight from the node v_k in VS_0 and using it as the root node, searching its parent up until v_1, v_2, v_3 to obtain VS_0^k , and traversing all of the nodes of v_k in VS_0 to obtain the set $VS_1 = \{VS_0^k | V_k \in VS_0\}$
3. Starting from VS_0^k in VS_1 , find each node in VS_0^k and loop through the steps in Step 2 to get $VS_2 = \{VS_1^k VS_1 | V_k \in VS_1\}$
4. Continue from VS_2 and repeat Step 3 until node $\overleftarrow{cdt}(v_i)$ in the searched sequential associated flight is not delayed, and the propagation is stopped

ALGORITHM 5: Delay reverse traceability algorithm.

Input: Node service time of flight $FA_{w_i}^j$ whose chock-off node is delayed (i.e., v_9 is delayed)

Output: The set $VS = \{VS_0, VS_1, VS_2, \dots, VS_p\}$ of propagated associated nodes starting from flight $FA_{w_i}^j$

1. Search inside the flight: starting with node v_9 of flight $FA_{w_i}^j$, identify its parent nodes v_3, v_7 , and v_8 to filter out the delayed nodes, and keep doing so until the search reaches the root nodes v_1 , and obtain the internal node association set VS_0
2. Search between flights: seeking the node $\overrightarrow{cdt}(v_k)$ in the postorder-related flight from the node v_k in VS_0 and using it as the root node, searching its children up until v_9 to obtain VS_0^k , and traversing all of the nodes of v_k in VS_0 to obtain the set $VS_1 = \{VS_0^k | V_k \in VS_0\}$
3. Starting from VS_0^k in VS_1 , find each node in VS_0^k and loop through the steps in Step 2 to get $VS_2 = \{VS_1^k VS_1 | V_k \in VS_1\}$
4. Continue from VS_2 and repeat Step 3 until node $\overrightarrow{cdt}(v_k)$ in the searched sequential associated flight is not delayed, and the propagation is stopped.

ALGORITHM 6: Delay forward propagation algorithm.

TABLE 3: Original records of flight ground service process data sample.

Key node name	Record value	Id	Service time	Service time
Flight arrival time	10:20	CA2335		
Flight departure time	11:20	CA2336	Ground service time	60 min
Arrival passenger vehicle	10:08	Pass vehicle 1	Passenger disembarks	11 min
Food vehicle	10:09	Food vehicle 2	Food service	16 min
Clean vehicle	10:09	Clean vehicle 3	Clean service	8 min
Water vehicle	10:09	Water vehicle 2	Water service	6 min
Arrival baggage vehicle	10:18	Baggage vehicle 1	Baggage loading service	14 min
Departure passenger vehicle	10:23	Passenger vehicle 2	Passenger boarding	17 min
Oil vehicle	10:14	Oil vehicle 1	Aircraft fueling	12 min
Departure baggage vehicle	10:31	Baggage vehicle 3	Baggage unloading	15 min
Flight chock-off time	11:08	—	—	—

The average daily flight time distribution is depicted in Figure 6, where 11:00 a.m. and 8:00 p.m. show obvious flight-intensive phenomena and the rest period displays a zigzag time distribution. We can evaluate the delays of these six vehicles according to the flight ground service time criteria that were published by the Civil Aviation Administration of China, since the arrival time and departure time of the flight were known. In addition, these service process nodes depicted in Figure 1 were used to determine the delay time in a certain way. Then, we discrete the original data according to the data processing approach in Section 4.1.1 to

transform the initial dataset for the VDT-BN and breakpoint simulation model. Table 4 exhibits the conversion sample data.

4.4.2. Experimental Setup. In terms of the experiments, we chose 80% of the datasets (148 days) for training our proposed model, which included the parameters of the VDT-BN and the grid parameters of the breakpoint simulation model. The remaining 20% of the datasets (38 days) was used for the model's validation analysis. Using the Unity3D platform as a foundation, the simulation model was created. The flight ground service

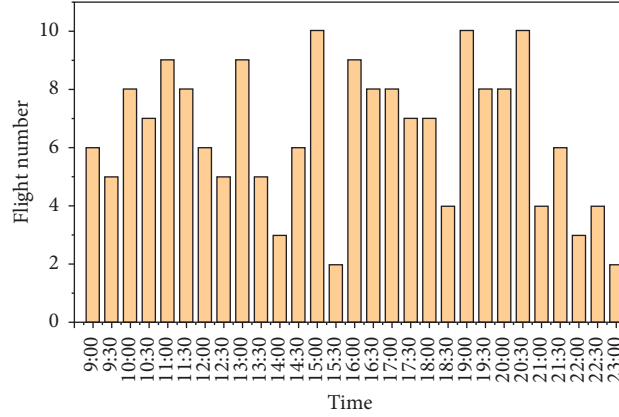


FIGURE 6: Density distribution of the airport flight number under 30-min time interval.

TABLE 4: Sample of discrete transformed flight ground service data.

Node name	Record value	Delay time	Discretized value	Node symbol
Arrival passenger vehicle	10:08	5	1	V1
Food vehicle	10:09	8	2	V4
Clean vehicle	10:11	0	0	V5
Water vehicle	10:12	3	1	V6
Arrival baggage vehicle	10:18	11	2	V2
Departure passenger vehicle	10:23	12	2	V7
Oil vehicle	10:14	0	0	V3
Departure baggage vehicle	10:31	6	2	V8
Flight chock-off time	11:08	4	1	V9

process is constructed through a multiagent model, and the simulation calculation and analysis are driven by real-time data.

We set up four sets of experimental scenarios, including the analysis of the VDT-BN diagnostic index, diagnostic algorithm performance comparison, breakpoint simulation model analysis, and flight delay propagation. The specifics were as follows:

- **Diagnosis model judgment analysis experiment:** In accordance with the real-time diagnosis index established in Section 4.1.3, the vehicle delay level and flight delay level are combined with the VDT-BN model to predict and assess the future situation of a flight delay, judge the accuracy of diagnosis, and assess the lowest diagnosis threshold, which is used as the foundation for delay propagation in the flight ground service digital twin system.
- **Comparative experimental analysis with VDT-BN diagnostic algorithms,** including decision tree model, random forest model, dynamic Bayesian model, and LSTM model. The decision tree model and random forest are based on historical flight service node data to forecast delays of future flights, and the dynamic Bayesian model treats the entire ground service network as a fixed structure, chooses to ignore the correlation between multiple flights, and performs serialization processing.
- **Breakpoint simulation accuracy analysis and evaluation experiment:** the service time distribution

parameters are learned and the breakpoint simulation model is combined to verify the prediction accuracy of flight ground service time in the real dataset.

- **Delay cause and propagation analysis experiments:** the causes of delays in the flight ground service process are analyzed using the reverse cable source algorithm and a detailed delay path diagram is obtained. Besides, when diagnosing the cause of the flight delay, we use the forward propagation algorithm combined with the simulation platform to analyze the future propagation path and guide the actual flight ground service operation management.

5. Experimental Analysis and Discussion

5.1. Diagnosis Model Judgment Analysis Experiment

5.1.1. Validity Examination of Diagnostic Indicators. Regarding the two diagnostic metrics: the average delay time of service vehicles Dc_{w_i} and the average delay time of flight ground service time Df_{w_i} , we randomly selected two typical days of data from the off-season where the volume of airport flights is relatively low and the peak season with a high volume for analysis.

As shown in Figures 7(a) and 7(b), multiflights are superimposed in w_i , and the X-axis represents a w_i interval according to the 30-min time interval. There are 28 divided time periods within the range of 9:30 a.m.–23:00 p.m. The same conclusion can be drawn from the two graphs: the ground service time for future flights is always delayed ahead

of time for vehicles. In other words, after a vehicle delay occurs first, it takes some time for it to spread before the flight delay occurs in the following interval, and the current interval sends a vehicle delay. Due to the service association of several vehicles and aircraft having some buffer time, there is also the possibility that the flight may not be delayed. This phenomenon consequently indicates that by diagnosing the delays of the service vehicles, we may evaluate the potential later flight ground service delays in advance warning.

5.1.2. Flight Delays' Diagnostic Analysis. Using the VDT-BN diagnosis algorithm, we predict future flight delays at intervals of $1w_t$, $2w_t$, ..., and $5w_t$ from the current interval, respectively, and compare those predictions with the actual delays to verify their accuracy. In total, we set 5 diagnosis lengths with a time range of 30 min–150 min.

The results are shown in Figure 8, where the line graphs in the upper half reflect the results of the five diagnostic lengths and the true values, respectively, and the bar graphs in the lower half represent the errors between the observed and predicted. We continue to use the 2 days of typical data that were chosen in Figure 8 according to five different lengths of diagnostic intervals. Figure 8(a) shows that there are flight ground service delays between the hours of 11:00 and 16:00. All five of the prediction lines identified flight delays, but the diagnostic results of the various lines differ in some respects. In Figures 8(a) and 8(b), similar phenomena can be observed.

Consequently, we can draw the following two conclusions: (1) There is no such thing as a practical flight sending a delay and then diagnosing it without delay; instead, all five diagnosis lengths are capable of detecting the delay trend. (2) The following error Figures 8(a) and 8(b) shows that as the forecast interval lengthens, the deviation gradually increases. The error lies in the $(-1.2, 1.6)$ interval, and the majority of the errors are more than 0, indicating that the predicted delays are worse than the real ones. (3) For real managers, who are more inclined to the former in terms of preliminary diagnostic results about whether there will be future delays and the severity of delays, this conclusion is acceptable from the point of view of real-time airport operating diagnostics. (4) By connecting real-time data, our following breakpoint simulation model provides a comprehensive evaluation of the potential propagation and the impact that occurs after delays occur.

5.1.3. Accuracy Statistical Analysis. We chose a 30-day dataset to conduct the statistical analysis, and we assessed the diagnostic accuracy of each time period using MAPE and MAE metrics. Where MAPE stands for the average error percentage between the diagnostic value and the true value, and MAE stands for the average absolute error. The smaller the value of the two indices, the more accurate the prediction. This experiment was performed to verify the diagnostic accuracy and stability of the method, and the prediction interval was increased from the previous $5w_t$ to $10w_t$, which means 300 min in the future.

Figure 9 displays the findings from our experiments, which reveal that when the diagnostic interval lengthens, the

error keeps growing. With an MAE of 0.12, the MAPE prediction accuracy for $1w_t$ is approximately 89% (MAPE value was 11%). As the length increases to $4w_t$, the accuracy drops to 69% (MAPE value was 31%) and for $6w_t$, it drops to 58% (MAPE value was 42%). Therefore, from the overall statistical analysis, the model in this work has a good diagnostic effect within 120 min (with about 80% diagnosis accuracy) and can provide early warning analysis for managers 2 h in advance, which is also better for airport managers to develop optimization strategies.

5.1.4. Minimum Threshold Analysis. Since this required minimum threshold was not specified during the previous analysis and testing, we performed the diagnosis once the service vehicle was delayed, which might potentially result in the phenomena of excessive warning during the actual application. As a result, it is essential to build the appropriate minimal diagnostic thresholds and conduct a correlation analysis between the diagnostic thresholds and flight delay indicators. To obtain appropriate diagnostic judgment thresholds, we count the association between the diagnostic thresholds and future delays in different w_t periods, as shown in equation (23), where $n_{wt}(Dc_{wt})$ indicates the value of the flight delay condition in w_t interval, and if there exists a flight delay in W_t , then $n_{wt}(Dc_{wt})$ is count as 1, and $prob(w_t)$ denotes the probability of flight delays occurring in the future w_t interval for a total of M counted numbers. A total of 100 sets of data were selected for the analysis, that is, $M = 100$, and the values of the diagnostic threshold ranged from 1.0 to 4.0 according to an increase of 0.2 to obtain 20 change intervals. The length of diagnosis for the future interval was chosen from $1w_t$ to $6w_t$. Then, the relationship between the diagnostic threshold and the future delays was calculated using equation (24).

$$n_{wt}(Dc_{wt}) = \begin{cases} 1, & \text{if } Df_{wt} > 0, \\ 0, & \text{else,} \end{cases} \quad (23)$$

$$prob(wt) = \frac{1}{M} \sum_{j=1}^M n_{wt}(Dc_{wt}). \quad (24)$$

Figure 10 illustrates that when the diagnostic indicator is at 1.2, the percentage of sending flight delays within 3 h ($6w_t$) is 20%, while at 2.6, the probability of future flight delays is 80%. As a result, in the subsequent digital twin system applications' scenery, we select the lowest diagnostic threshold of 1.2 and the highest is 2.6. These diagnostic index findings are crucial for effective operation management as they indicate that delays are less likely to happen below 1.2 and more likely to happen after exceeding 2.6.

5.2. Performance Comparative Analysis of Different Diagnostic Models. To further validate the efficiency of the VDT-BN diagnostic algorithm, we performed a comparative analysis with the decision tree model, dynamic BN, random forest model, and LSTM model. The DBN model,

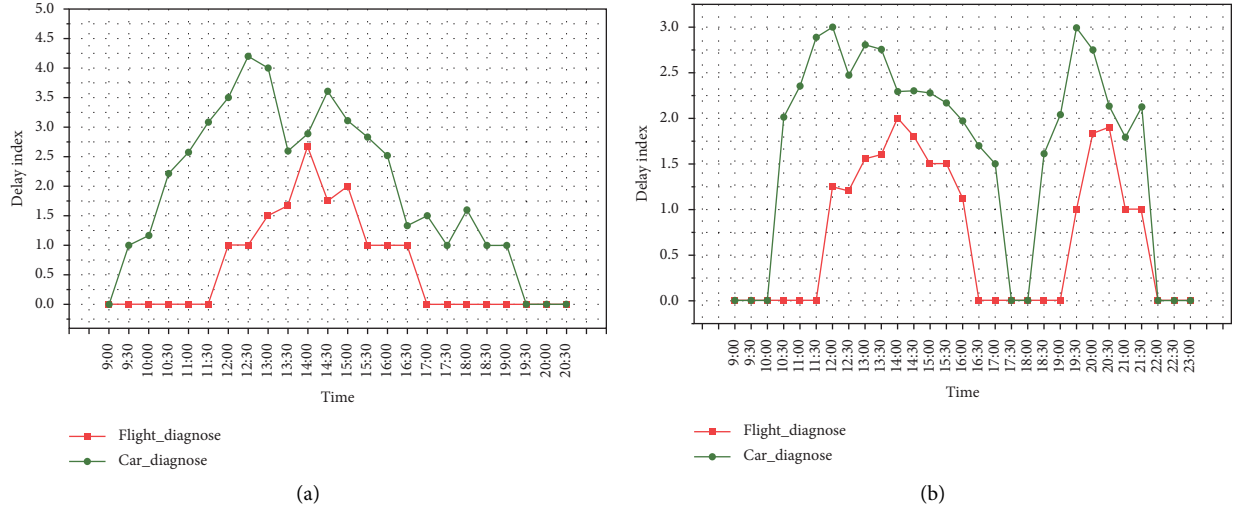


FIGURE 7: Aircraft and service vehicles delay diagnostic result analysis for a typical day in low and high season. (a) June 17, 2019, off-season real diagnostic results. (b) July 15, 2019, peak season diagnostic results.

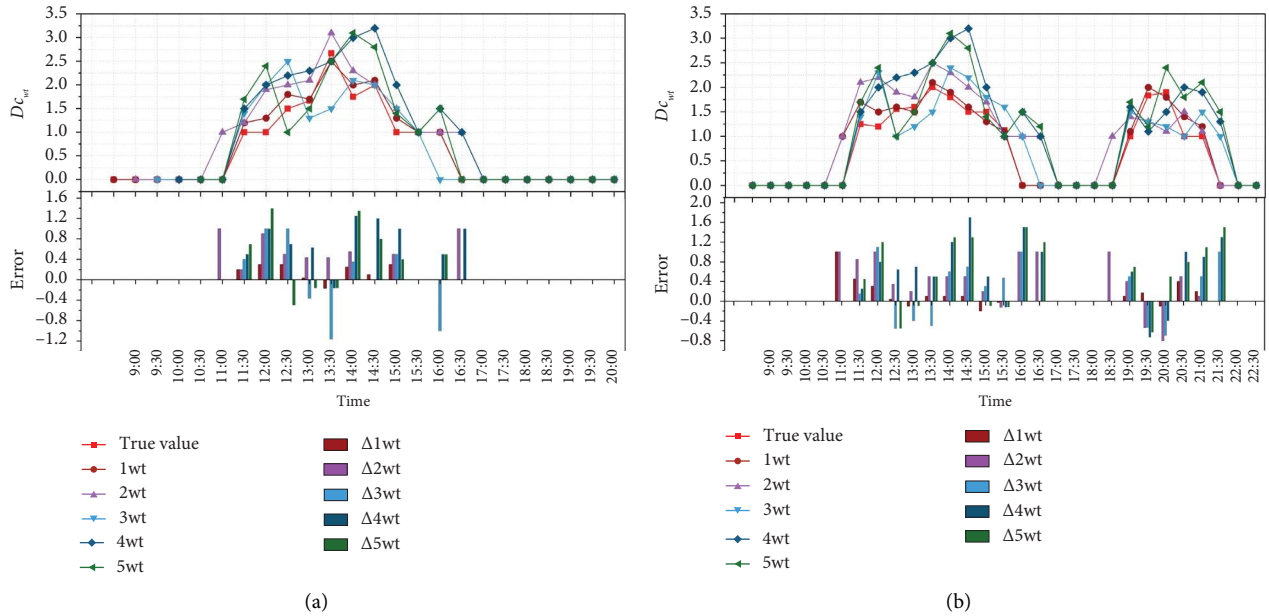


FIGURE 8: Performance analysis of different diagnostic lengths on peak and off-season. (a) Off-season. (b) Peak season.

for example, takes into account the node relationships in the flight ground service process, while the other 3 models are employed by building feature variables. The dynamic BN is an expansion of the BN with the inclusion of an adjacent connectivity network. When making a predictive diagnosis, we estimate each flight in turn based on the spited w_t interval, and eventually the service node time of each flight is obtained. The random forest, decision tree, and LSTM models, on the other hand, forecast the service node time based on the data in the previous interval $4w_t$. The business process association between flights is then learned in a black-box manner via the model's feature learning capability. As in Section 5.1, we chose two

measures, MAPE and MAE, and a 30-day dataset to compare the average MAPE with the MAE.

Table 5 shows the results of the five algorithms. We can obtain the following conclusion: (1) The random forest and decision tree models outperform DBN and LSTM in terms of both short-term and long-term diagnostic accuracy, and our proposed VDT-BN shows the best performance. (2) Since the LSTM model fails to identify the business process relationship between flight data, via feature extraction of the time sequence in the flight ground security node data, so its diagnostic prediction precision accuracy is the lowest, especially after expanding to $3w_t$, where its MAPE value is about 75%, which is equivalent to only 25% accuracy. (3) The DBN model

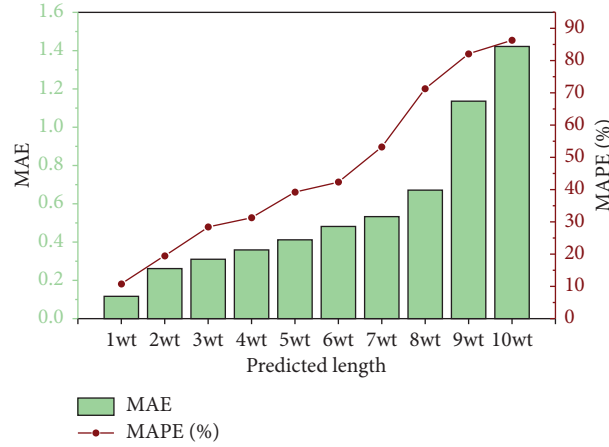


FIGURE 9: Performance evaluation results of average MAPE and RMSE on different diagnostic lengths.

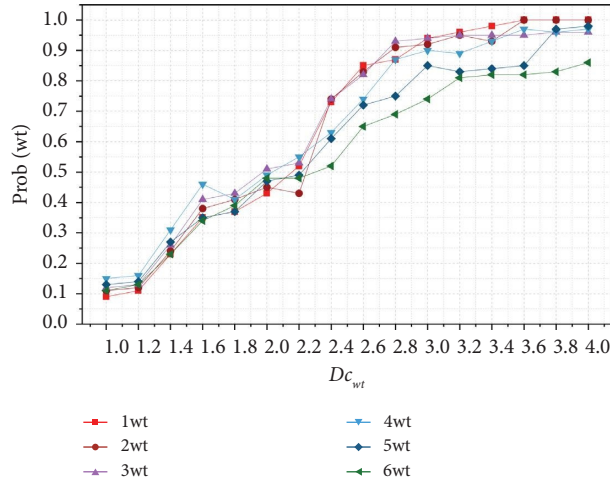


FIGURE 10: Diagram of the relationship between the values of diagnostic indicators and the probability of flight delay.

incorporates the graph structure interaction between flight ground protection nodes and its diagnostic precision is improved when compared to the LSTM model; nevertheless, when the prediction interval is extended, its precision remains poor. In the $3w_t$ interval, for example, the MAPE value of DBN is 71.35%, making it difficult to meet actual demand. This could be owing to the complicated network relationship formed by multiflights, which is difficult to depict using the typical DBN network structure, resulting in low diagnostic accuracy. (4) Both the random forest and decision tree models performed well in terms of short-term diagnostic results, with MAPE values of 18.42% and 14.36%, respectively, which are near to the MAPE assessment index of 10.75% of our proposed VDT-BN model. However, when the diagnostic interval lengthens, the error of the random forest and decision tree models grows.

Overall, our proposed VDT-BN model achieves the best diagnostic performance. This may be due to the fact that we were able to extract the features between these data more effectively than other methods that used fixed network structure or black-box-type learning techniques by combining the correlation relationship between multiple flights with the delayed path relationship of the service vehicles.

5.3. Accuracy Analysis of the Breakpoint Simulation Model.

We learn the grid parameters using historical datasets with a total of 6 types of vehicles and 8 service nodes conducted. We fit the distributions using the methods in Section 4.2.1 and use the Kullback–Leibler divergence to evaluate how well these distribution models fit the actual service time. Figure 11 shows the fitting results of the optimal 5 distributions. We can find that clean vehicle follows a normal distribution, while food vehicle service time gamma distribution is better than a normal distribution. Luggage vehicles of departing passengers and arriving passengers follow beta distribution and exponential distribution, respectively; baggage vehicle loading and unloading service obey Cauchy distribution and beta distribution; and finally, oil vehicles and water vehicles obey beta distribution.

After obtaining these node service time's distribution results, we integrated with the business of flight ground service process in a breakpoint simulation model to experiments by calculating the node time of service vehicles as well as the flight chock-off time to verify the accuracy of the model prediction performance, and we still choose MAPE and MAE as the evaluation index. For the statistical analysis,

TABLE 5: MAE and RMSE performance of the algorithms with VDT-BN under different diagnostic lengths.

Length	Metrics	LSTM	DBN	Random forest	Decision tree	VDT-BN
1_{wt}	MAE	0.37	0.29	0.25	0.21	0.12
	MAPE%	31.32	26.45	18.42	14.36	10.75
2_{wt}	MAE	0.51	0.38	0.34	0.31	0.26
	MAPE%	44.57	36.15	29.61	27.83	19.45
3_{wt}	MAE	0.83	0.74	0.52	0.43	0.31
	MAPE%	76.31	71.35	51.34	41.05	28.43
4_{wt}	MAE	1.05	0.81	0.53	0.61	0.36
	MAPE%	81.25	73.89	53.27	63.42	31.27
5_{wt}	MAE	1.14	0.87	0.64	0.71	0.41
	MAPE%	83.14	79.64	65.21	68.36	39.25
6_{wt}	MAE	1.16	0.93	0.79	0.73	0.48
	MAPE%	82.67	73.56	69.42	70.15	42.37

Note: Bold values indicate that this is the most effective comparison.

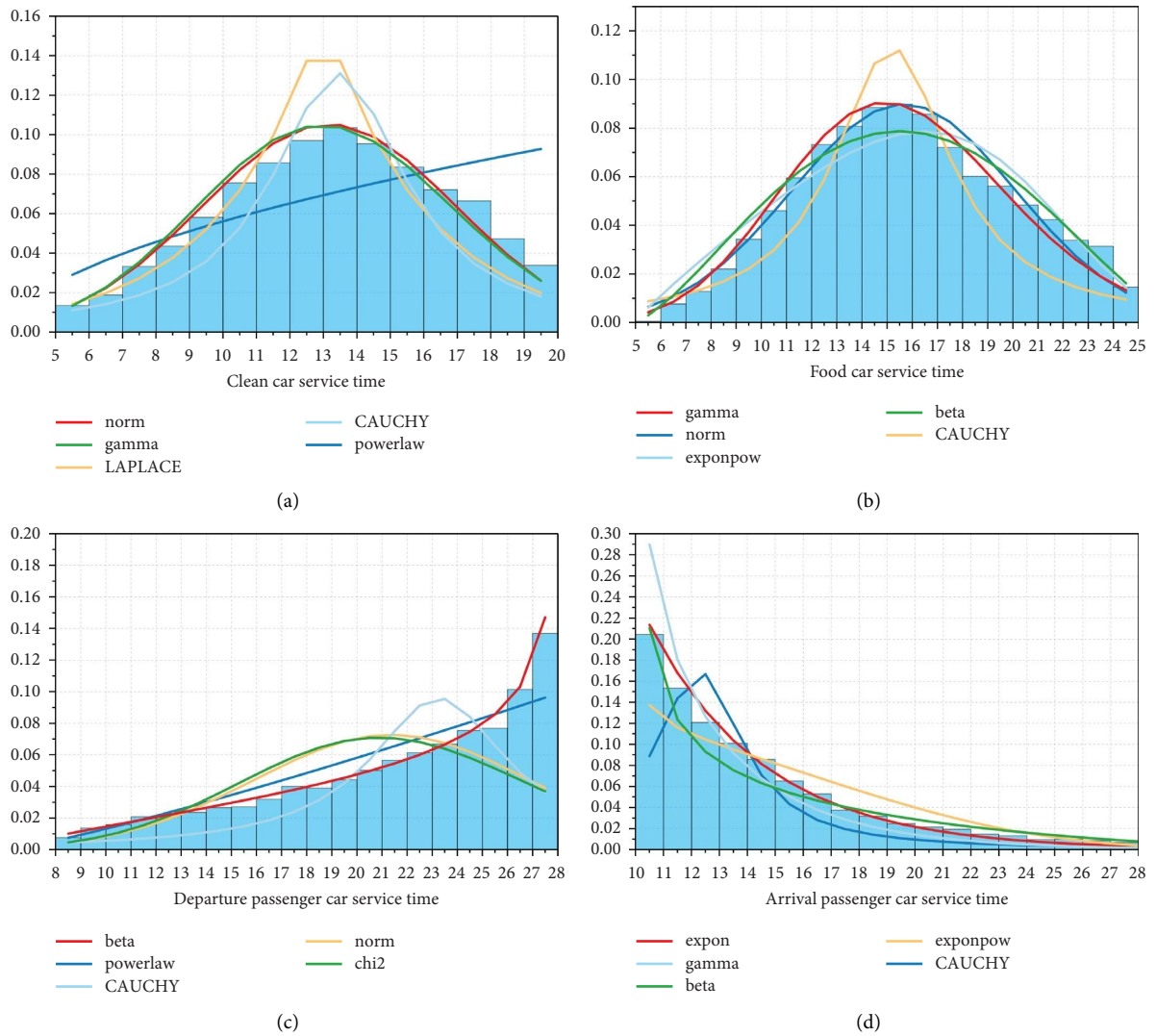


FIGURE 11: Continued.

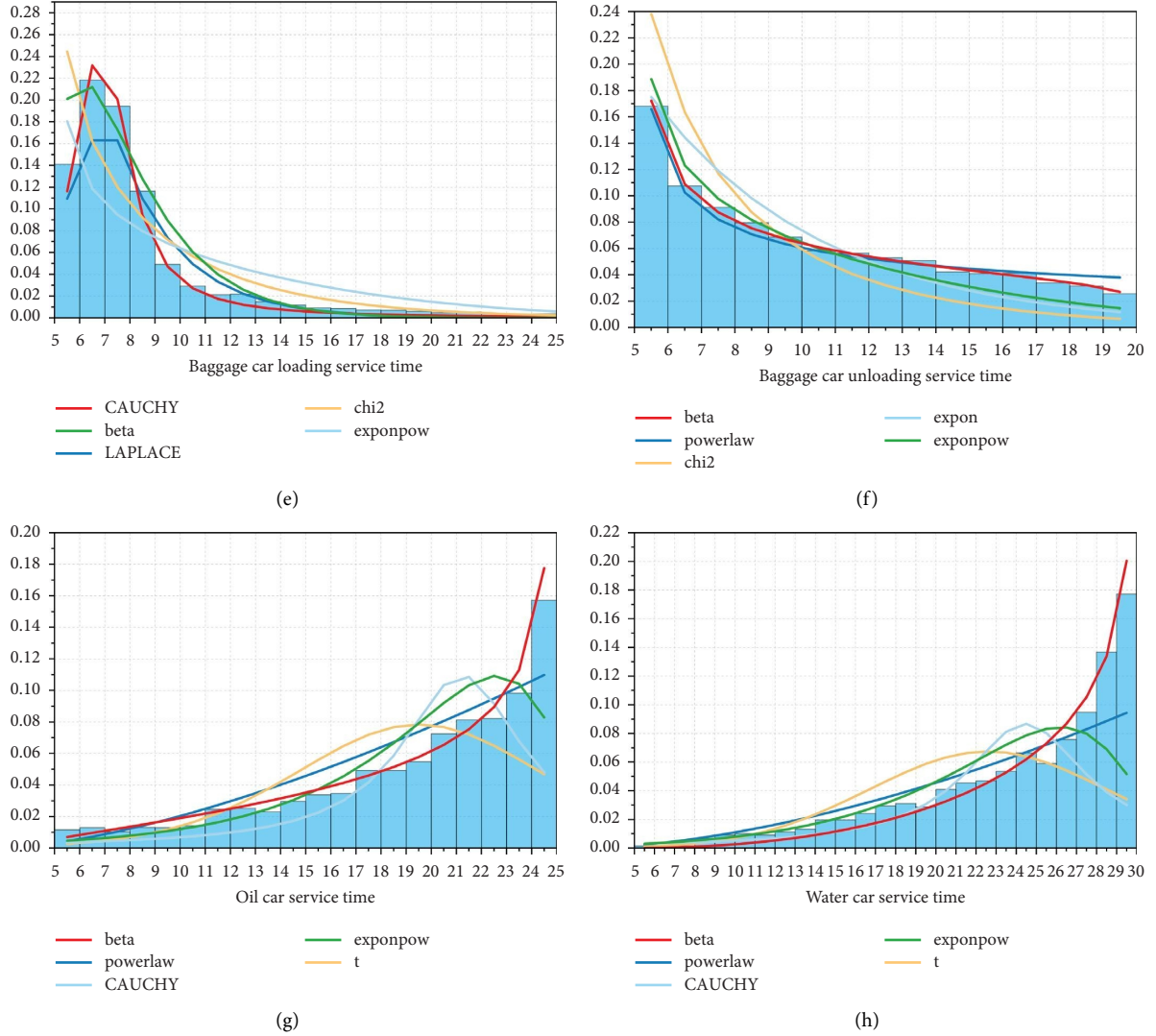


FIGURE 11: Estimated results of service time distribution of different ground service nodes. (a) Norm distribution. (b) Gamma distribution. (c) Beta distribution. (d) Exponential distribution. (e) Cauchy distribution. (f) Beta distribution. (g) Beta distribution. (h) Beta distribution.

a total of 30 days' flight datasets were used to obtain the average MAE and MAPE values.

Figure 12 shows the evaluation results. According to Figure 12(a), the average error of all types of vehicle service nodes is less than 14 min, and the prediction time of the chock-off node is about 13.2 min. According to the airport's flight ground service time standard and delay requirement, our prediction accuracy can meet the real demand within 15 min. Furthermore, the MAPE values in Figure 12(b) are all less than 0.2, indicating that we can achieve 80% prediction accuracy, which may serve as the basis for subsequent propagation and cause mining analysis of flight ground service delays.

5.4. Delayed Contribution and Propagation Path Analysis. In this section, we analyze the contributing paths that resulted in flight ground service delays as well as the propagation paths after the delays occurred. The real flight datasets are shown in

Figure 13 along with multiflight relationships, and it can be seen that the flight ground service delay begins in the w_3 interval. By using the VDT-BN diagnostic algorithm, we are able to predict subsequent flight delays and perform early warning analysis when the delay indicator is sent in w_1 . If there is no connection between two flights, it means that the service vehicle is not associated with these two flights and is not in the forward or reverse propagation association. The red box denotes the flight with the chock-off node delayed. Using flight $FA_{w_5}^2$ as our starting place, we apply the delay tracing and propagation technique in this study to identify the root reasons for the delay in the flight's chock-off node and assess how the delay has affected the following flights.

5.4.1. Delayed Contribution Analysis. The issues that delayed flight $FA_{w_5}^2$ are depicted in Figure 14 along with their causes and paths of propagation. The delay propagation path of the $FA_{w_5}^2$ to the future aircraft is depicted in Figure 15.

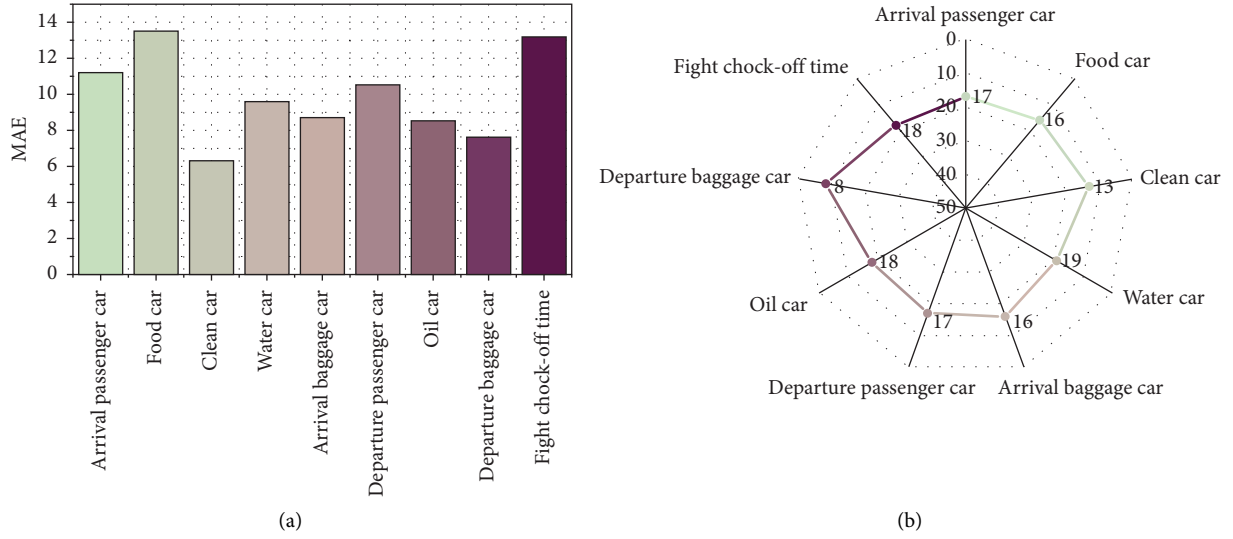


FIGURE 12: Accuracy evaluation results of the breakpoint simulation model. (a) MAE evaluation value of different service nodes. (b) MAPE evaluation value of different service nodes.

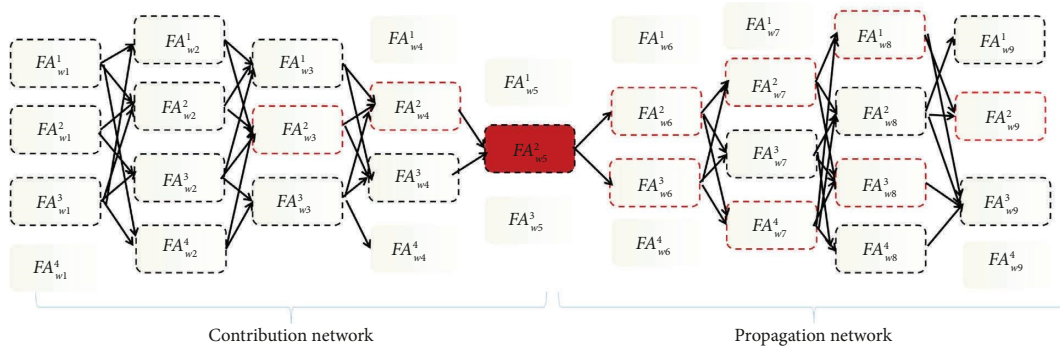
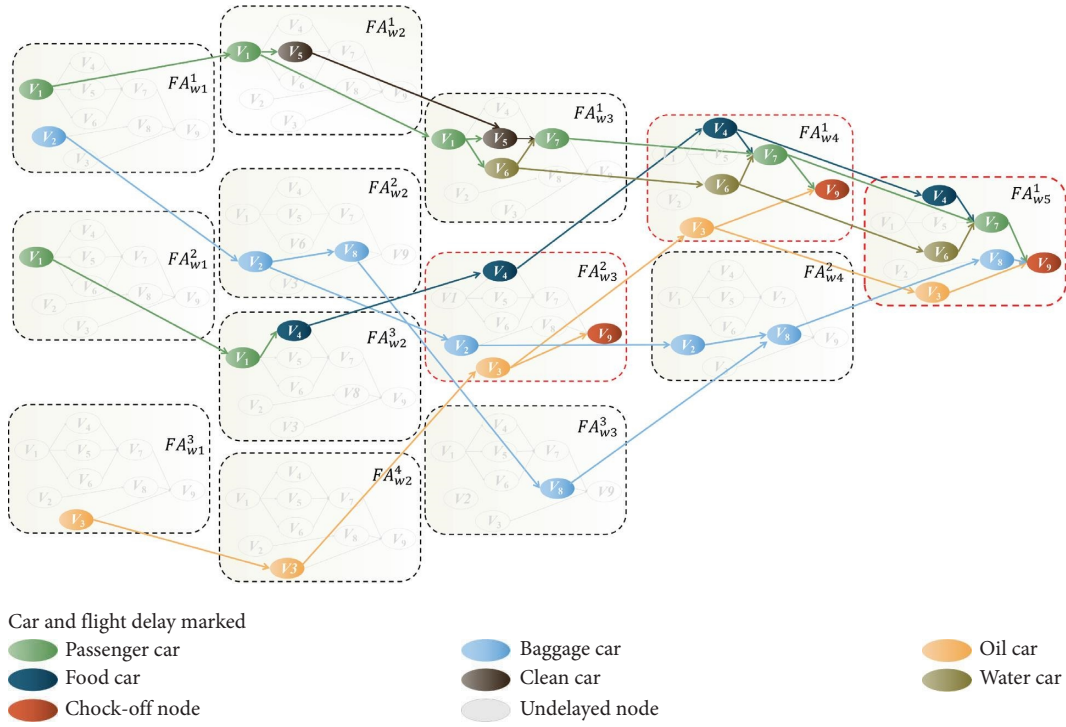
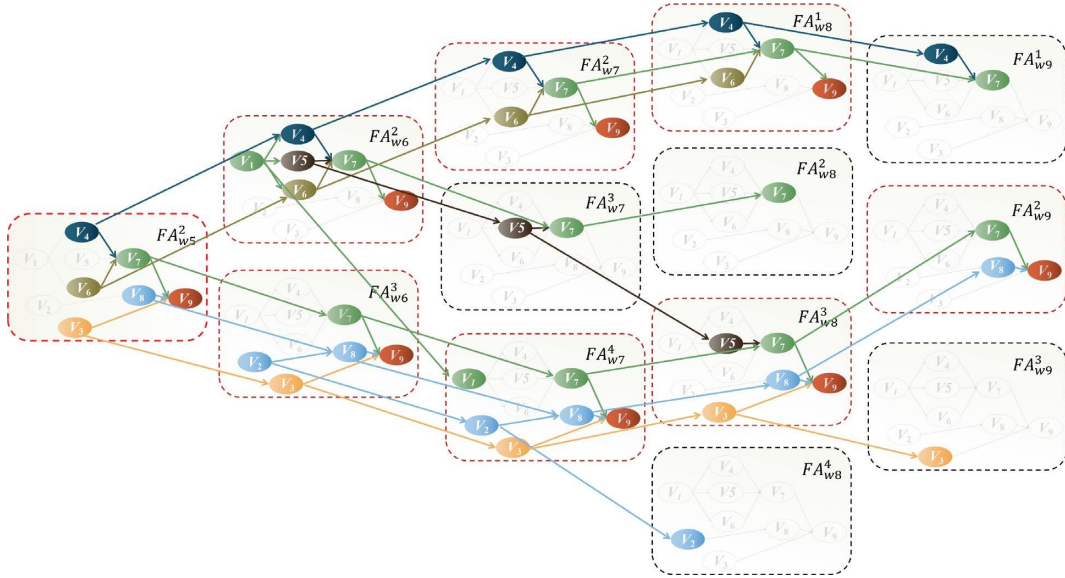


FIGURE 13: Delay contribution and propagation paths of flight $FA_{w_5}^2$, where the red box denotes the flight with the the chock-off node delayed, the black dashed box indicates the flight with service vehicle delay but no delay in the chock-off node, and the rest indicates the normal flight.

According to Figure 14, we can find that the five delayed nodes v_3, v_4, v_6, v_7 , and v_8 (oil vehicle, food vehicle, clean vehicle, water vehicle, and departure passenger vehicle) are the key factors that contribute to the delay of the flight chock-off node in $FA_{w_5}^1$. We then identify the underlying cause by tracing backwards the associated flights using our proposed Algorithm 4. For instance, v_4 node in flight $FA_{w_5}^1$ can obtain the reverse contribution path $FA_{w_5}^1(v_4) \leftarrow FA_{w_4}^1(v_4) \leftarrow FA_{w_3}^2(v_4) \leftarrow FA_{w_2}^3(v_4) \leftarrow FA_{w_1}^2(v_1)$, that is, the delay of the v_1 node on aircraft $FA_{w_1}^2$ is the initial cause of the delay of the v_1 node on flight $FA_{w_2}^3$, which then goes to the v_1 node and finally causes the delay of the v_4 node on flight $FA_{w_5}^1$. Finally, the root nodes in flight $FA_{w_1}^1(v_1), FA_{w_1}^1(v_2), FA_{w_1}^2(v_1), FA_{w_1}^3(v_3)$ is detected by the reverse contribution paths of each node in flight $FA_{w_5}^1$, and the four contribution paths caused by these four root nodes can be represented by C1, C2, C3, and C4. Therefore, airport

managers can effectively implement better control strategies for the flight ground service process by analyzing the propagation paths of these delay contributors.

5.4.2. Delayed Propagation Analysis. After identifying these warning messages through the VDT-BN model, we need to further analyze how these delays propagate. Through the breakpoint simulation model in the digital twin system, we can simulate and analyze the situation after the delay, and then use our proposed Algorithm 5 to analyze their propagation path. Following the diagnosis of the delay in flight $FA_{w_5}^1$, the detailed propagation network is shown in Figure 15. We obtained the propagation pathways of each node sequentially after identifying the five nodes v_3, v_4, v_6, v_7 , and v_8 in flight $FA_{w_5}^1$, and recognize five forward propagation paths, denoted by P1, P2, P3, P4, and P5. For instance, node v_6 , which was delayed in flight $FA_{w_5}^1$, propagated to node v_7 , and together with node v_8 , caused the delay of the v_9 node.

FIGURE 14: Detailed contribution path of ground service time delay in flight FA_{w5}^1 .FIGURE 15: Detailed propagation path of ground service time delay in flight FA_{w5}^1 .

The propagation route of this node was $FA_{w5}^1 (v_6) \rightarrow FA_{w6}^2 (v_6) \rightarrow FA_{w7}^2 (v_6) \rightarrow FA_{w8}^1 (v_6) \rightarrow FA_{w8}^1 (v_7)$, and this node's delay ultimately ended the propagation in node v_7 of flight FA_{w8}^1 . Besides, the delays in the three flights $FA_{w6}^2, FA_{w7}^2, FA_{w8}^1$ all eventually contributed to delays in the v_9 node.

5.4.3. Control Strategy Analysis. Using the forward propagation and the reverse contribution path analysis, we can

quantitatively depict the propagation routes P1–P5 following a flight delay and the paths C1–C5 that cause flight failures, which is essential for developing an effective flight ground service control strategy.

As shown in Tables 6 and 7, we put together these paths involving the relationship of the service vehicle in different intervals and analyze different control strategies. For instance, the C1 route in Table 6 shows that only one node v_1 of the first flight begins to be delayed. Once this delay reaches the w_2 interval, it spreads to two further nodes, v_1 and v_4 ,

TABLE 6: Number of vehicle delays for the four contributing paths in different intervals.

Zone	Contribution path vehicle delay number				Total vehicle delay number	Flight chock-off delay number
	C1	C2	C3	C4		
w_1	1	1	1	1	4	0
w_2	2	2	2	1	7	0
w_3	4	2	1	1	8	1
w_4	2	2	1	1	6	1
w_5	2	1	1	1	5	1

TABLE 7: Number of vehicle delays for the five propagation paths in the different intervals.

Zone	Contribution path vehicle delay number					Total vehicle delay number	Flight chock-off delay number
	P1	P2	P3	P4	P5		
w_5	1	1	1	1	1	5	1
w_6	1	3	2	1	2	9	2
w_7	1	3	3	1	2	10	3
w_8	1	3	2	1	2	9	2
w_9	1	2	0	1	1	5	1

and continues until the w_5 interval. In order to minimize the maximum 4 service vehicle delays at the time of w_3 , we need first to improve C1-affected service vehicles. In addition, the propagation pathways in Table 5, P1 and P4, denote v_3 node and v_8 node, i.e., oil vehicle and passenger vehicle, respectively. They have obvious one-way propagation characteristics, i.e., each diffusion range is not wide and will only affect one service vehicle. In order to minimize their delays, we may control how such vehicles are laid out in advance. For P2 and P3 routes involving v_4 and v_6 nodes, namely, the food vehicle and clean vehicle, we need to focus on the on-time control of such vehicles to prevent the delay propagation effect in the network. These service vehicles usually have shorter service times and serve more flights, especially when there are more flights in one interval.

5.5. Discussion. Based on the comprehensive validation through four distinct experimental aspects, we have demonstrated the efficacy of the proposed digital twin system application algorithms in this work. By conducting experimental analyses on real airport datasets, we verified the parameter thresholds for delay diagnosis and evaluated them based on practical airport validation experiences. Subsequently, based on these parameter analysis results, we established multiple comparative algorithms to confirm the effectiveness of the two proposed delay diagnosis indicators. The comparative experiments demonstrated that the VDT-BN diagnostic algorithm proposed in this work can better capture the probabilistic relationships among multiple flight ground service processes compared to traditional decision tree models, random forest models, dynamic Bayesian models, and LSTM models, thus enabling more timely and accurate delay diagnoses. Following this, we conducted verification and analysis of the breakpoint simulation model, covering the simulation results of eight service stages during the flight ground service process. We validated the simulation accuracy through fitting methods and analyzed the effectiveness of the breakpoint simulation

model by predicting the final time. Finally, we integrated these diagnostic and simulation methods, using the digital twin platform for real-time diagnostic warnings and delay propagation analysis, which yielded effective delay tracing paths and delay propagation paths. By comparing these with the actual flight ground service processes, we confirmed the effectiveness of delay propagation and tracing. Based on these analyses, we can provide timely warning information to flight ground service personnel and offer real-time delay propagation paths after delays occur, effectively addressing the issues of insufficient auxiliary decision-making information and inadequate judgment capabilities for airport operation management.

Furthermore, through the analysis of these four experiments, we have comprehensively applied the L3-level digital twin framework proposed in this work. According to the three-tier structure of the framework, data mapping layer, function execution layer, and application service layer, we realized an L3-level digital twin capable of generating auxiliary decision-making applications for airport operations based on real-time data. If new application scenarios need to be added in the future, this framework still possesses versatility; it allows for the addition of more data in the mapping layer, updates to algorithms at the function execution layer, and allows the provision of more simulation services using the breakpoint simulation model. Ultimately, application functional logic can be implanted in the application layer. Therefore, the entire framework has a certain degree of versatility.

However, our approach still confronts several challenges in practical applications, such as data missing and dynamic changes in the flight ground service business processes. These factors could potentially significantly diminish the theoretical diagnostic accuracy and effectiveness of the algorithms proposed in this paper. Therefore, we should further consider the realities of operational data, incorporating methods to impute missing data within the digital twin platform. In addition, to enhance the resilience

of our diagnostic algorithms, we need to design the existing BN structure to be more flexible, enabling it to adapt to the complexities and variability of real-world business operations.

6. Conclusions

Early diagnosis of the flight ground service process and identification delay propagation process are of great significance to improving airport operation efficiency. In this work, the digital twin technology is applied to delay diagnosis and propagation analysis, which solves the limitation of traditional methods on insufficient early diagnosis and imprecise simulation prediction. Compared to the existing digital twin applications in the airport field, such as L1- and L2-level 3D visualization asset management and resource scheduling plan maintenance, our developed digital twin system reached a higher L3-level maturity.

Moreover, we present a general digital twin framework primarily divided into three parts: the data transmission tier, the twin functional tier, and the application service tier. The data transmission tier serves as a unified platform for data integration, while the twin functional tier offers real-time diagnostic models. The application service tier, on the other hand, provides delay propagation analysis models. Looking ahead, we anticipate the potential to extend this framework by incorporating additional data sources, thereby enabling enhanced business management capabilities and further applications. The main result was as follows: (1) we propose a VDT-BN delay diagnosis model to portray the dynamic dependencies of the whole airport flight ground service process and realize timely and effective diagnosis of flight delays. (2) The experimental results show that VDT-BN has about 90% early warning accuracy at 30 min and still has about 69% accuracy after expanding to 90 min. (3) To solve the problem of low accuracy of traditional offline simulation methods, we propose a breakpoint simulation method that supports real-time simulation of any diagnostic warning moment, and the experiment shows that the breakpoint simulation model has more than 80% prediction accuracy. (4) Based on the diagnostic warning and real-time simulation data, we propose a delay traceability and propagation algorithm to analyze the causes and propagation paths of flight delays, which can more accurately formulate the control strategy of the ground protection operation process.

The broad benefits of the digital twin-based delay diagnosis and propagation approach for airport flight ground service processes are as follows: (1) By synchronizing airport real-time operation and presenting 3D visualization in the digital twin platform, airport managers can monitor the global situation more intuitively. (2) Using diagnostic algorithms to identify flight delays, and combining them with delay traceability algorithms to identify the key factors that cause flight delays, effectively improve the timeliness and accuracy of airport operation control. (3) The delay forward propagation algorithm can identify specific flight and vehicle association paths, which provides effective support for ground operation managers to further control future delays in real time.

However, it is found from the case study that the proposed method has limitations in fulfilling the entire functions of delay diagnosis and propagation application: (1) The digital twin system is highly dependent on the real-time data collection from the physical airport, and there may be some missing data or data anomalies, which will affect the accuracy of our delay diagnosis result. (2) Since the factors influencing flight delays at airports include not only local airport ground service process but also macroweather and factors from other airports, this work mainly studies the delay factors caused by local airports.

Considering these limitations, we should further implement data governance techniques in digital twin systems to improve the quality of real-time data so that machine learning applications can get more accurate results. In addition, we also need to consider more factors in the real-time breakpoint simulation model, such as flight ground operation information from other airports and weather factors, to achieve a more accurate analysis of flight delays. Finally, with the advancement of sensor technology and the widespread adoption of unmanned and minimally manned devices, our proposed framework can be further extended to other areas, such as cargo terminals and airport terminals. Using this idea of delay diagnosis and early warning as well as propagation analysis and cause mining and control strategy analysis, we can further improve the intelligent operation of airports.

Data Availability Statement

The data used to support the findings of this study are confidential, nonpublic data.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Chang Liu: methodology, software, investigation, writing original draft, and funding acquisition. YuanYuan Zhang: methodology, validation, conceptualization, writing, review, and editing. YanRu Chen: methodology, conceptualization, writing, review, editing, supervision, and funding acquisition. ShiJia Liu: methodology, validation, writing, review, and editing. ShunFang Hu: resources, methodology, and project administration. Qian Luo: resources, methodology, and project administration. LiangYin Chen: conceptualization, review, editing, and supervision. Chang Liu and YuanYuan Zhang contributed equally to this work.

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